The Block Universe

A Scientific and Philosophical Exploration of Spacetime and Physical Reality

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Introduction

This manuscript grew out of a personal intellectual curiosity about physics and, more specifically, about the nature of physical reality. I am not a physicist, nor do I have the formal educational background necessary to develop or propose new physical theories. My academic and professional background is in engineering, which provides some useful analytical tools, but I approach the subjects discussed here as an amateur enthusiast seeking to understand questions that I find deeply interesting.

Of particular interest to me are the *ontological* questions underlying our descriptions of the physical universe. What does it mean for a physical event to exist? What is the nature of time? What does it mean for the universe to have a complete physical history? And how should we understand our own existence as conscious observers embedded within that reality?

Modern physics provides extraordinarily successful mathematical descriptions of the natural world. Yet a mathematical framework that accurately predicts observations does not necessarily provide an intuitively satisfying picture of what physical reality *is*. The distinction between predicting what we observe and understanding what those observations imply about the nature of reality is the principal motivation for this manuscript.

My objective is to develop a coherent way of thinking about several foundational topics in physics within a common ontological framework. Rather than treating questions about spacetime, the flow of time, quantum measurement, causality, and free will as entirely separate problems, I want to explore whether they can be understood from a small number of shared principles.

The guiding approach is to adopt as few additional postulates as possible, favoring those that appear logically reasonable and compatible with established physical laws. The hope is that a relatively simple ontological perspective can provide a more unified and intellectually satisfying interpretation of phenomena that otherwise appear disconnected or paradoxical.

The Block Universe interpretation of spacetime serves as the central organizing perspective for this exploration. Beginning with the implications of relativity, the manuscript examines how a four-dimensional description of physical reality might inform our understanding of time, physical laws, quantum mechanics, conscious experience, free will, and causal consistency.

The objective is interpretive rather than predictive. I do not claim to derive new physical laws, propose a replacement for existing theories, or produce new experimentally testable predictions. Nor do I suggest that ontological reasoning alone can establish the physical truth of a particular interpretation. The aim is more modest: to explore whether a coherent interpretation of physical reality can be constructed that remains consistent with what established physics tells us, while making as few additional assumptions as reasonably possible.

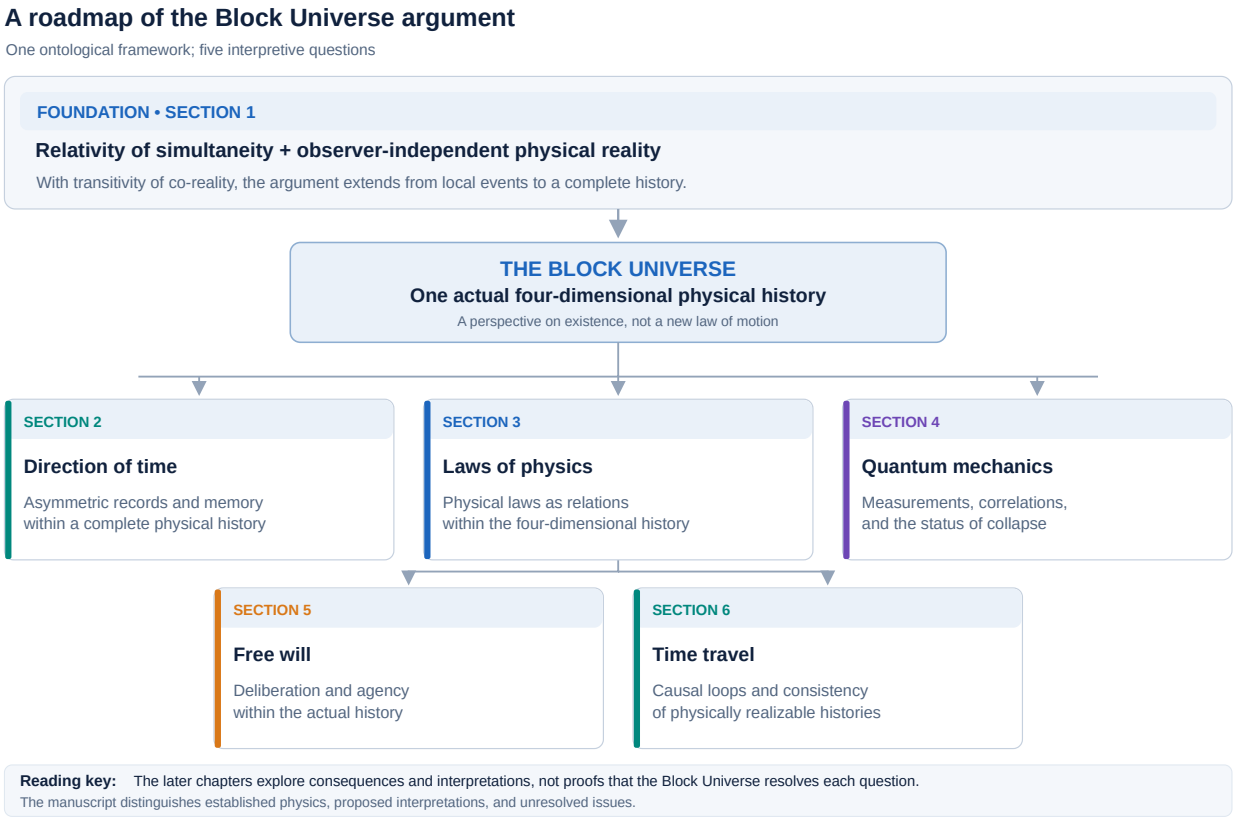


Figure 0.1. *Roadmap of the Block Universe argument.*
Relativity motivates the ontological framework; the subsequent sections examine its scope and limitations across five foundational topics.

Many of the ideas discussed in this manuscript have been explored extensively by physicists and philosophers. I make no claim that the underlying concepts or arguments are original. What I hope to achieve is a synthesis that is personally meaningful: a connected set of arguments that I can revisit, question, and refine as my understanding develops.

The manuscript was developed through extensive interactive discussions with ChatGPT. I used these conversations to explore ideas, challenge assumptions, construct thought experiments, examine mathematical arguments, and fill gaps in my knowledge of physics. ChatGPT also assisted substantially in organizing and drafting the text. The resulting manuscript is therefore a collaborative product of my questions, reasoning, and interpretations, together with AI-assisted explanations, analysis, and writing. This assistance should not be mistaken for independent scientific validation, and any technical claims remain subject to verification against established literature.

Throughout this exploration, I have tried to distinguish between what established physics demonstrates, what a particular interpretation suggests, and what remains speculative or unresolved. Logical self-consistency is an essential requirement, but it is not sufficient to establish physical truth. Where the proposed framework cannot explain something, I would rather acknowledge that limitation than introduce unsupported assumptions simply to preserve an appealing picture.

Ultimately, this manuscript is written primarily for myself. It is a record of an ongoing attempt to understand the universe and our place within it. Its value to me lies not in whether it contributes something new to science, but in whether it provides a coherent intellectual framework that I can return to, reconsider, and improve.

If it also proves interesting or useful to another curious reader, that would be a welcome additional outcome.

1. From Relativity to the Block Universe

Central claim of this section: Special relativity establishes that there is no observer-independent global present for spatially separated events. The relativity of simultaneity allows the same physical event to be assigned to the past, present, or future of different inertial observers. We begin with the reality of an event for an observer located where it occurs and introduce a single ontological principle: **the physical existence of an event is independent of the observer's location or reference frame.** Applying this principle consistently to the spacetime relationships of special relativity, and recognizing the transitivity of physical co-reality, leads to a four-dimensional reality in which past, present, and future events belong to the same physical history. This is the Block Universe. It is not assumed at the outset, but derived from special relativity together with the stated ontological premise.

1.1 Newtonian Absolute Time and the Universal Present

In classical Newtonian mechanics, time is treated as an absolute, universal parameter: a single temporal ordering that, in principle, applies identically to all observers regardless of their location or state of motion. Within this framework, the universe can be pictured as progressing through a sequence of three-dimensional spatial states, each associated with the same universal time.

This picture naturally supports *presentism*, the philosophical view that only the present moment exists, while the past no longer exists and the future does not yet exist. It is important, however, to distinguish this philosophical interpretation from Newtonian mechanics itself: Newtonian physics supplies an absolute time parameter, but it does not by itself constitute a complete philosophical argument for presentism.

1.2 Einstein's Spacetime: From Separate Space and Time to Minkowski Geometry

Special relativity replaces Newton's absolute time with a spacetime structure in which spatial and temporal coordinates depend on the inertial frame used to describe events. There is no privileged inertial frame from which a universal temporal slicing of the universe can be defined.

Observers in relative motion can disagree about spatial distances and temporal durations. More importantly for the question of *now*, they can disagree about whether two spatially separated events are simultaneous.

Although the individual quantities Δx and Δt vary between inertial frames, the spacetime interval remains invariant:

$$\Delta s^2 = -c^2\Delta t^2 + \Delta x^2 + \Delta y^2 + \Delta z^2$$

Here c is the invariant causal speed of relativity. In ordinary units it also relates temporal and spatial coordinates: multiplying a time interval by c gives a quantity with units of length. This permits time to be represented geometrically alongside the spatial coordinates as ct, x, y, z.

The resulting Minkowski spacetime is a four-dimensional geometric structure. An object's history can be represented within it by a **worldline**: a continuous curve through spacetime. This geometrical description does not by itself require the additional philosophical claim that time literally “does not pass”; rather, it provides a framework in which an object's entire spacetime history can be represented as a single geometric object.

1.3 What Does “Now” Mean?

The word “now” can refer to several different things, and separating them is essential to understanding relativity.

1. **Observed now:** the events whose light signals are currently reaching an observer. For distant objects, this is affected by the finite propagation time of light.
2. **Relativistic simultaneity:** the distant events that an observer's inertial coordinate system assigns the same time coordinate as the observer's current event. These events form a hypersurface of constant time in that frame.
3. **Ontological existence:** the philosophical question of whether events in the past, present, and future exist in the same sense. Special relativity does not, by itself, settle this metaphysical question.

The crucial point: The relativity of “now” concerns the second meaning. Two observers in relative motion can disagree about which distant events are simultaneous with their common local event. They are not merely receiving different visual information; their inertial coordinate systems assign different simultaneity relations to spatially separated events.

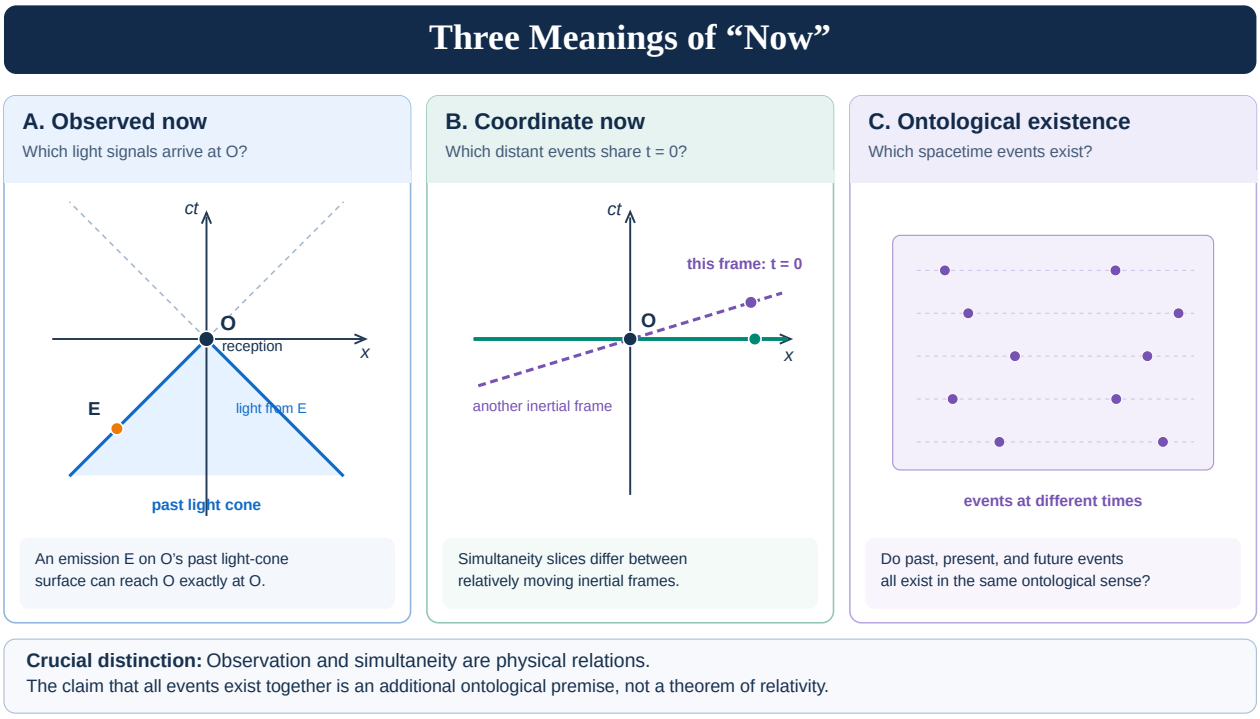


Figure 1.1. “Now” can refer to arriving information, coordinate simultaneity, or a claim about existence.
Past-cone boundary = signals received at O; simultaneity depends on reference frame; existence requires interpretation.

1.4 The Relativity of Simultaneity

Figure 1.2 is a two-dimensional representation of Minkowski spacetime. A complete representation of Minkowski spacetime is four-dimensional, with three spatial dimensions and one time dimension, and therefore cannot be directly depicted in an ordinary two-dimensional diagram. For clarity, the figure suppresses two of the spatial dimensions and displays only one spatial dimension together with the time dimension. This simplification does not affect the generality of the argument: the relativistic relationships illustrated in the diagram are properties of four-dimensional Minkowski spacetime, and the conclusions drawn from them apply to the full 4D spacetime description.

Consider three inertial observers who pass through the same local event at the origin of a Minkowski diagram:

- **Observer A:** at rest relative to the unprimed coordinate system, with v = 0.
- **Observer B:** moving at a positive relativistic speed v' relative to A.
- **Observer C:** moving at a still greater speed v'', where v'' > v'.

For an inertial frame moving at speed v relative to A, the Lorentz transformation is

$$t' = \gamma(t - vx/c^2), \qquad x' = \gamma(x - vt), \qquad \gamma = 1/\sqrt{(1-v^2/c^2)}$$

The key feature is the term vx/c^2 in the transformation of time. The time coordinate assigned to a distant event depends on both its time coordinate and its spatial position in the original frame.

For an event simultaneous with the origin according to the moving observer, t' = 0. Therefore:

$$t' = 0 \;\Rightarrow\; t = vx/c^2$$

Thus, except at x = 0, the events that B considers simultaneous with the origin do not generally satisfy t = 0 in A's frame. This is the mathematical expression of the relativity of simultaneity.

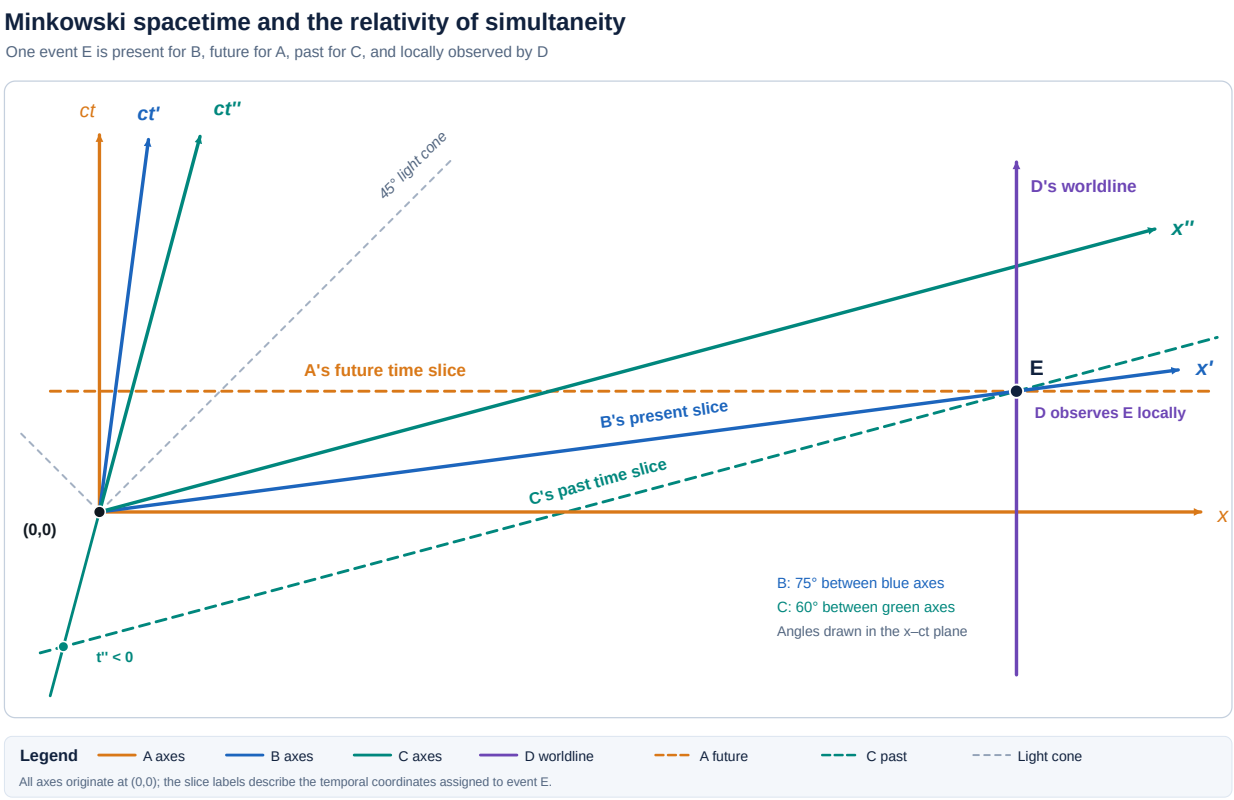


Figure 1.2. *Minkowski spacetime diagram illustrating tilted axes and different hypersurfaces of simultaneity for inertial observers in relative motion.* Event E is simultaneous with B's origin (t' = 0), but is in A's future (t > 0) and C's past (t" < 0). Grey dashed lines mark the 45° light cone. The vertical coordinate is ct, with matching coordinate scales for x and ct; the diagram is schematic.

Diagram Analysis

1. **Reference-frame axes:**
   - Observer A uses the unprimed x and t coordinates.
   - Observer B uses the primed x' and t' coordinates.
   - Observer C uses the double-primed x" and t" coordinates.
   - Observer D is local to event E (explained in later subsections).

2. **Coordinate distortion:**
   Relative motion changes the coordinate decomposition of spacetime. In a Minkowski diagram whose vertical coordinate is ct, the time axes and simultaneity axes appear tilted relative to one another. Greater relative velocity produces a greater tilt, approaching the light-cone directions as v approaches c.

3. **Hypersurfaces of simultaneity:**
   - **Orange horizontal dashed line:** A's positive constant-t (future) time slice.
   - **Blue x' axis:** B's t' = 0 (present) time slice.
   - **Green inclined dashed line:** C's negative constant-t" (past) time slice.

1.5 Event E: How the Same Distant Event Can Be Future, Present, or Past

Now consider a distant event E, separated from the common origin by a spatial distance $d > 0$. Choose E so that it is **spacelike separated** from the origin. This qualification is essential: only spacelike-separated events can have their temporal ordering or simultaneity relation changed by a change of inertial frame.

Suppose the velocity parameters are chosen so that E lies on B's hypersurface of simultaneity. Then:

1. **Observer B:** Event E satisfies $t' = 0$. B therefore assigns E the same coordinate time as the origin. In the precise relativistic sense, E is **simultaneous** with B's present event.
2. **Observer A:** The same event E has $t > 0$. A therefore assigns E a later time than the origin: E lies in A's **future**.
3. **Observer C:** For a sufficiently different relative velocity, the same event can have $t'' < 0$. C therefore assigns E an earlier time than the origin: E lies in C's **past**.

Nothing contradictory has happened. The three observers are using different inertial coordinate systems to describe the same two spacetime events. The disagreement arises because simultaneity for spatially separated events is not invariant.

What is—and is not—relative: For spacelike-separated events, observers can disagree about which happens first or whether they are simultaneous. For timelike- or lightlike-separated events, the causal ordering is invariant: all inertial observers agree on which event can causally precede the other.

1.6 Why There Is No Universal “Now”

The Event E example reveals the central conceptual consequence of special relativity. If A, B, and C are all inertial observers and none is physically privileged, there is no frame-independent rule that identifies one particular distant hypersurface as *the* present moment of the universe.

In other words, the following question has no observer-independent answer within special relativity:

> **“Which events happening at distant locations are occurring right now?”**

A can give one answer. B can give another. C can give another. Their answers are not different because one observer has made a measurement error. They arise from the Lorentz transformation and the geometry of Minkowski spacetime.

This is the precise sense in which the “now” is relative.

Relativity of “now”:
There is no single, observer-independent three-dimensional hypersurface of distant events that all inertial observers must identify as the present. Each inertial frame defines its own hypersurfaces of simultaneity.

1.7 From Local Reality to Observer-Independent Existence

Consider a physical event E and an observer D located at E when it occurs. For D, the event is locally real. No inference about distant events, delayed signals, or philosophical interpretation is required to establish this starting point.

Now introduce another observer, B. Suppose initially that B and D coincide at E. Since they encounter the same local event, there is no question that E is physically real for both.

Now imagine, instead, that B is located at an infinitesimally small spatial separation from E, on the same simultaneity hypersurface in an inertial frame. This is a comparison of hypothetical observer locations, not a description of B physically traveling from E.

We ask a simple question: Does E cease to exist merely because B is no longer located at E?

We adopt the principle that it does not. The physical reality of an event is independent of which observer considers it.

The separation between B and D can be increased, and the construction repeated across arbitrarily small increments. In this way, local physical reality is extended across spatially separated events without requiring a privileged observer.

Special relativity now supplies the decisive additional ingredient: the relativity of simultaneity.

By changing B's inertial frame, an event E that was simultaneous with B's present can instead be assigned to B's coordinate past or future, provided the two events are spacelike separated. Nothing about E changes under this transformation. Only the temporal coordinates assigned to it change.

If E is physically real for D, and physical existence is observer-independent, E cannot cease to exist for B merely because B's frame assigns it to the past or future.

The local argument rests on the following ontological principle:

The physical reality of an event does not depend on the identity of the observer or the inertial reference frame used to describe it.

This is a claim about physical existence, not merely about knowledge. It is stronger than saying that B can later establish that E occurred. The claim is that the physical status of E does not depend on who describes it. At this stage, the argument concerns local and spacelike-separated comparisons; the extension to events throughout spacetime is established in Section 1.8 using transitivity of co-reality.

There is an important restriction: the temporal order of timelike- or lightlike-separated events cannot be reversed by a Lorentz transformation. The argument does not claim otherwise. To establish the result for events separated in time as well as space, we must extend the construction beyond a single pair of spacelike-separated events.

1.8 Constructing the Four-Dimensional Block Universe

Consider two events P and Q anywhere in Minkowski spacetime, including events that are timelike separated. Even if P and Q cannot be simultaneous in any single inertial frame, we can connect them through intermediate events that are spacelike separated from their neighbors.

To see this, use a one-dimensional spatial coordinate for simplicity. Let P have coordinates (t = 0, x = 0), and let Q have coordinates (t = T, x = 0). Choose an intermediate event R with coordinates (t = T/2, x = L), where L is sufficiently large that L > c|T|/2. Then both P–R and R–Q are spacelike separations. Each pair can be simultaneous in some inertial frame, although the two frames need not be the same. Spatial translations and rotations, or the same construction with additional intermediate events, extend this reasoning to arbitrary pairs of events in Minkowski spacetime.

The physical significance is that no single simultaneity slice has to contain the entire chain. Each adjacent spacelike pair can be treated using the local-reality construction of Section 1.7. If co-reality is transitive—if P is real together with R, and R is real together with Q, then P and Q belong to the same physical reality—the construction connects events across the full four-dimensional spacetime.

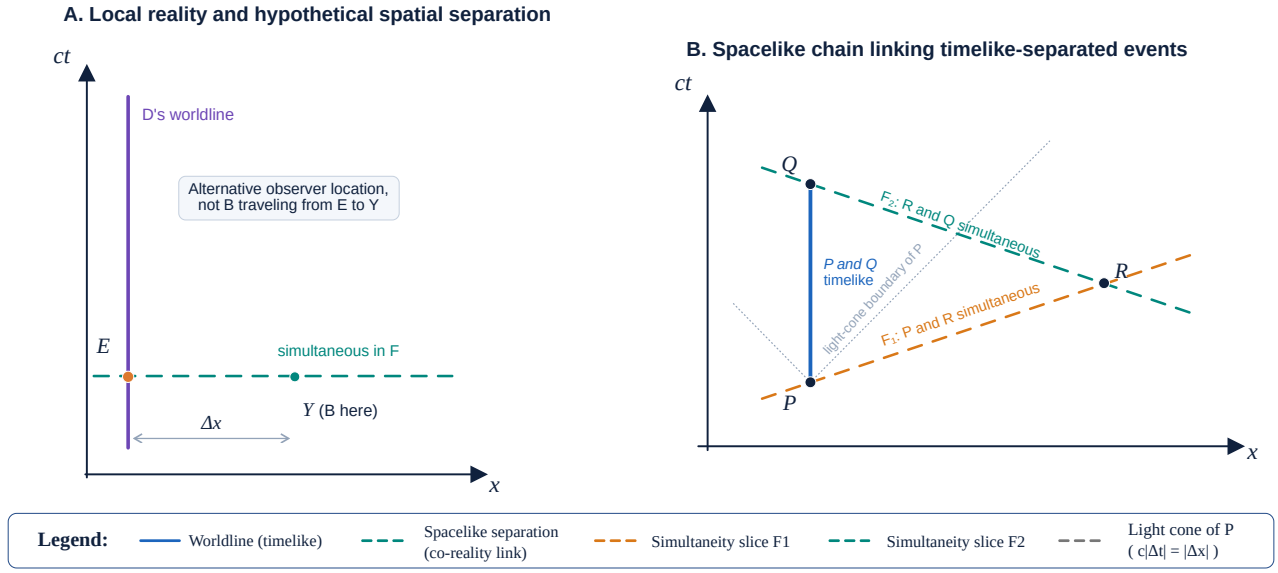


Figure 1.3. *The extension from local physical reality to a complete four-dimensional history.* (A) An event E on observer D's worldline and a nearby event Y, occupied by observer B, spacelike-separated from E and simultaneous with E in a suitable frame. (B) A spacelike chain P – R – Q in which P and Q are timelike-separated, while P – R and R – Q are spacelike-separated. By connecting such pairs and *adopting transitivity of co-reality, the construction extends to a complete four-dimensional history.*

This transitivity is not an additional dynamical law. It makes explicit what is meant here by one observer-independent physical reality: reality is not partitioned into mutually incompatible sets that depend on which observer or simultaneity frame is chosen. Nevertheless, it is an ontological commitment, and it should not be mistaken for a mathematical theorem of the Lorentz transformations alone.

We can now state the reasoning in its complete form:

1. A physical event E is locally real for an observer D at E.
2. We adopt observer-independent physical reality for the spacelike-separated comparisons of Section 1.7: a change of observer or inertial frame does not change the physical reality of E.

3. Special relativity makes distant simultaneity frame-dependent. A spacelike-separated event can be labeled past, present, or future relative to B's local event without changing the event itself.
4. Chains of spacelike-separated events, together with transitivity of co-reality, connect arbitrary events, including those separated by timelike intervals.
5. Therefore, physical existence is not restricted to one advancing three-dimensional present. The events of the actual history form a four-dimensional physical reality: the Block Universe.

The spacelike-chain construction and transitivity of co-reality now support the stronger conclusion:

If an event E is physically real for its local observer D, then E is also part of physical reality for any other observer B, independently of B's spacetime location or inertial reference frame.

The Block Universe is thus the conclusion of the construction, not its starting assumption. Special relativity supplies the spacetime geometry; the premise of observer-independent existence, understood to include co-reality and its transitivity, supplies the ontological bridge.

A presentist can reject that bridge, or introduce a preferred foliation defining an objective present. Such a foliation is logically conceivable, but special relativity does not identify one. The disagreement is therefore over ontology, not over the Lorentz transformation or the observed relativity of simultaneity.

Here, *static* does not mean that matter, fields, or geometrical relationships remain unchanged from one time coordinate to another. It means that the actual four-dimensional history does not require an external time in which the universe as a whole advances from one present to the next.

1.9 General Relativity: Curved Spacetime and Dynamical Geometry

Special relativity is sufficient for the argument developed above for the Block Universe. The discussion of general relativity is included here for completeness, to show how the same four-dimensional spacetime framework is extended when gravity is taken into account. General relativity introduces the curvature of spacetime, in contrast to the flat spacetime of special relativity, as the geometric description of gravity. The arguments made so far, however, do not depend on spacetime being flat: they concern the four-dimensional nature of spacetime and the observer-dependence of temporal relations. Accordingly, the essential conclusions extend from flat Minkowski spacetime to curved spacetime as described by general relativity.

General relativity extends the geometric description from flat Minkowski spacetime to curved spacetime, represented mathematically by a four-dimensional manifold $(M, g_{\mu\nu})$. The gravitational field is encoded in the spacetime metric, and the metric is dynamically related to matter and energy by the Einstein field equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$$

Unlike the flat geometry of special relativity, the geometry of general relativity need not be static. The Einstein field equations are dynamical equations, and particular solutions can describe evolving gravitational fields, gravitational radiation, expanding or contracting cosmological geometries, and other time-dependent phenomena.

Nevertheless, individual physical histories can still be represented within a four-dimensional spacetime geometry. Worldlines of matter and trajectories of light are curves in that geometry, while the geometry itself is determined by the gravitational field equations together with appropriate initial or boundary conditions.

Thus, general relativity preserves and deepens the geometric lesson of special relativity without requiring the claim that spacetime as a whole is universally static.

1.10 Summary: From Relativity to the Block Universe

1. Newtonian mechanics permits a universal simultaneity relation through its absolute time parameter; presentism is a possible, but not compulsory, interpretation of that framework.
2. Special relativity replaces absolute simultaneity with frame-dependent simultaneity within Minkowski spacetime.
3. For spacelike-separated events, inertial observers may disagree about which event is earlier or whether the events are simultaneous. Timelike and lightlike causal ordering remains invariant.
4. We begin with a locally real event E and adopt the ontological premise that its physical existence is independent of the observer's location and inertial frame. This is stronger than merely saying that E occurred or that information about E may later arrive.
5. Applying this premise to relativity of simultaneity shows that E's physical existence cannot depend on whether another observer assigns it to the coordinate past, present, or future.
6. Chains of spacelike-separated events and the transitivity of physical co-reality extend the construction to the actual four-dimensional history, including timelike-separated events.
7. On this stated ontological premise, the resulting physical reality is the Block Universe. Special relativity alone does not prove the premise, but it provides the structure through which the conclusion is derived.

Core conclusion: The Block Universe need not be postulated at the outset. Begin with the reality of a local physical event, require that its existence be observer-independent, and apply that requirement consistently to the spacetime relations of special relativity. With the transitivity of co-reality made explicit, the result is a single four-dimensional physical history rather than an observer-independent present that advances through otherwise nonexistent events.

2. The Block Universe and Direction of the Flow of Time

2.1 From the Existence of Time to the Flow of Time

Section 1 established the Block Universe as a four-dimensional spacetime in which past, present, and future events all physically exist. The entire history of the universe can therefore be regarded as a single four-dimensional entity extending across the temporal as well as spatial dimensions. For the purposes of this discussion, we will take this block to encompass the complete history of the universe, extending back toward the Big Bang and its associated initial singularity. Every event contained within the spacetime—including events that we describe as past, present, or future—statically exists as part of the complete four-dimensional structure.

This raises a different question. If the entire spacetime exists as a whole, and time is simply one of the coordinates of that spacetime, why do we experience time as flowing in a particular direction?

The distinction between **time** and the **flow of time** is essential here. Within the Block Universe, the temporal coordinate provides an ordering of events just as spatial coordinates provide an ordering of locations. Nothing in the existence of a temporal coordinate by itself requires spacetime to physically move from one time to another. There is therefore no need to introduce a fundamental substance or process that causes the universe to advance through time.

We may therefore distinguish three related but different concepts:

- **Time as a coordinate:** the temporal dimension used to locate events within spacetime.
- **Temporal ordering:** the ordering of events along a timelike worldline, such as the sequence of physical states experienced by an observer.
- **The experienced flow of time:** the perception by a physical observer that it is progressing from one state to another, with a definite distinction between past, present, and future.

The first two are structural properties of spacetime and physical systems within it. The third is a property of an observer embedded within spacetime. The Block Universe therefore does not need to *flow* through time in order for an observer within the block to experience a flow of time.

In this sense, the flow of time is not a fundamental property of spacetime itself. It is a description of the experience of physical systems as they occupy successive states along their worldlines.

The Block Universe does not flow through time; observers experience a flow of time within the Block Universe.

2.2 The Observer as a Physical System in the Block

To understand this experienced flow, consider an observer as a physical system embedded within the four-dimensional spacetime of the Block Universe. The observer has a worldline extending through spacetime, and at different points along that worldline its physical state is different.

Let the physical state of the observer at three successive values of its temporal coordinate be represented by:

(T_1, S_1) (T_2, S_2) (T_3, S_3), $T_1 < T_2 < T_3$.

All three states are part of the Block Universe. Nothing about the block itself requires us to declare that the sequence is physically flowing from S_1 to S_2 to S_3. The reverse ordering can equally be described geometrically by traversing the same worldline in the opposite coordinate direction.

At this stage, therefore, we have established temporal order but not yet the *direction of experienced time*.

The question becomes: **what physical property of the observer distinguishes one direction along its worldline as the direction of experienced time?**

2.3 The Information Structure of Successive Brain States

For a human observer, the relevant physical system is the brain. The human brain is itself an entity within the four-dimensional spacetime of the Block Universe. As the brain's worldline extends through spacetime, the physical state of the brain changes from one point to another.

Suppose that the brain is in state S_2. That state contains physical information resulting from its interaction with the world at earlier points along its worldline. The brain may contain memories of previous experiences, records of previous events, learned information, physical modifications resulting from earlier interactions, and many other forms of information that are consequences of earlier states.

Consequently, the state S_2 contains information about aspects of S_1. The later state therefore carries physical records of the earlier state.

$S_1 \rightarrow S_2$: S_2 contains information about S_1.

By contrast, the state S_2 does not ordinarily contain corresponding information about the future state S_3. The brain at S_2 does not ordinarily contain memories or physical records produced by events that have not yet occurred along its worldline.

$S_2 \rightarrow S_3$: S_2 does not ordinarily contain information about S_3.

This creates an information asymmetry between the two directions along the observer's worldline. One direction is associated with states that have already left information in the current state; the other is associated with states that have not yet done so.

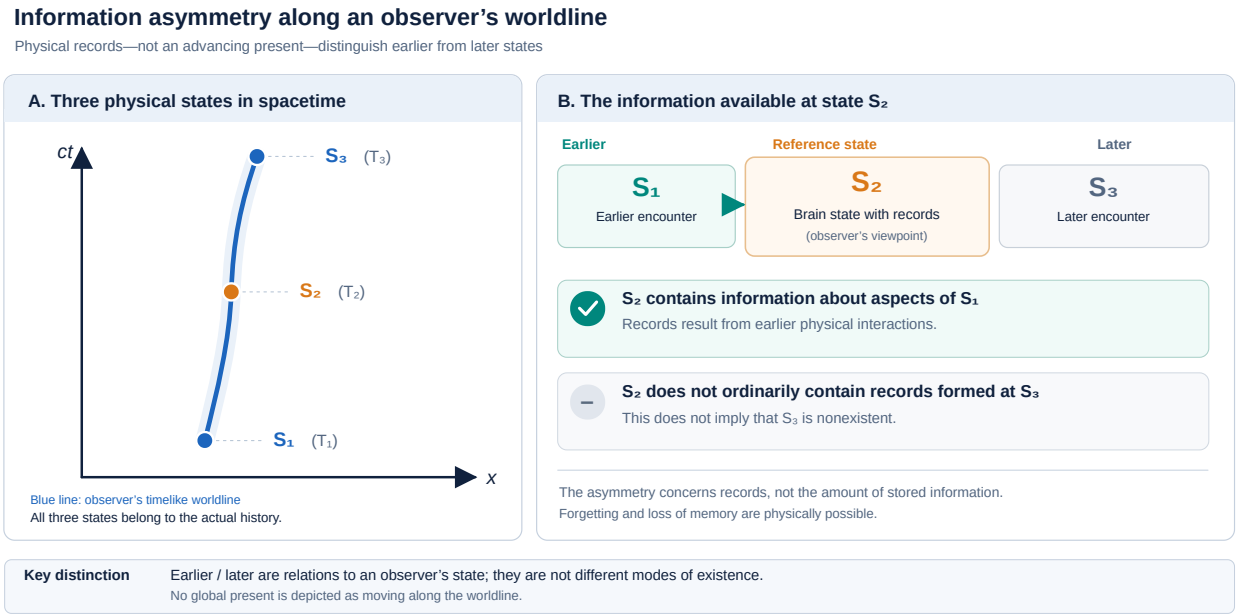


Figure 2.1. *Information asymmetry along an observer's worldline.*
The physical state S_2 contains records arising from earlier interactions, including aspects of S_1, but does not ordinarily contain records produced at S_3. All three states are parts of one four-dimensional history. The asymmetry provides a physical basis for distinguishing past from future without requiring time itself to flow or memories to accumulate without loss.

This asymmetry provides a physical basis for the observer's distinction between past and future. If S_2 is the brain's present state, then S_1 is experienced as part of the **past** because information about it is represented in the present state of the brain. Conversely, S_3 is experienced as part of the **future** because information about it is not yet represented in the brain's present state.

The experienced distinction can therefore be represented schematically as:

Past $\leftarrow S_1 \mid S_2 \mid S_3 \rightarrow$ **Future**

with S_2 representing the observer's present physical state.

The important point is that this distinction does not require the future to be nonexistent. S_3 already exists as part of the Block Universe. What distinguishes it from S_1 from the perspective of the brain is not its ontological status, but the information contained in the brain's current physical state.

Past, present, and future are therefore not necessarily three different modes of existence within the Block Universe. They are three different relationships between an observer's present physical state and other regions of spacetime.

2.4 The Direction of Experienced Time

We can now identify what is meant by the *direction* of the flow of time for an observer.

The brain does not merely occupy a sequence of physical states. Its successive states contain records of events associated with preceding states. The direction in which these records accumulate establishes an asymmetry along the brain's worldline.

At S_2, the brain contains information about S_1, while it does not ordinarily contain information about S_3. The brain therefore experiences S_1 as past and S_3 as future.

In this sense, the direction of experienced time can be associated with the direction of the observer's worldline along which physical records of earlier states are established in later states.

It is important, however, not to equate this simply with the statement that the brain must monotonically accumulate information. A brain can forget information, lose memories, or undergo local decreases in stored information. The relevant property is more fundamental: **the causal-information structure of successive physical states is asymmetric.** Later states ordinarily contain records and consequences of earlier states, while earlier states do not contain corresponding records produced by later states.

The direction of experienced time is therefore not defined by the absolute quantity of information stored in the brain. Rather, it is defined by the asymmetric orientation of information and physical records along the observer's worldline.

2.5 The Thermodynamic and Gravitational Origin of the Information Asymmetry

The existence of an information asymmetry in the brain raises a further question. Why should physical records be oriented in this particular temporal direction at all? Why should later physical states contain records of earlier states rather than the reverse?

This is where the thermodynamic history of the universe enters the argument. The universe possesses a profound thermodynamic asymmetry. The early universe was in an extraordinarily low-entropy state, while entropy has generally increased through the subsequent evolution of the universe. This provides an intrinsic physical asymmetry within the Block Universe itself.

The unusually low-entropy condition of the early universe is often referred to as the *Past Hypothesis*. It is important to recognize that the Block Universe does not explain why this particular boundary condition exists. However, this is not a difficulty unique to the Block Universe. Conventional dynamical descriptions of time likewise require an explanation for the special initial conditions that give rise to the thermodynamic arrow. The Block Universe does not seek to derive those conditions; it describes a complete four-dimensional physical history in which the resulting entropy gradients, irreversible processes, and formation of physical records are already present. The origin of the low-entropy boundary condition remains an open explanatory question, independent of whether time is understood as flowing or as a dimension of a static spacetime.

However, the low-entropy condition is not the whole physical story. The early universe was extremely homogeneous and isotropic on large scales, but it was not perfectly homogeneous. Small primordial density fluctuations were present. These fluctuations provided the seeds for gravitational structure formation: under the influence of gravity, initially small differences in density could grow, eventually producing galaxies, stars, planets, and the complex physical environments in which chemistry, biology, information processing, and observers could arise.

This point is especially important because gravity changes the way we think about entropy. A highly smooth matter distribution can correspond to very low gravitational entropy, whereas gravitationally clumped structures—most dramatically, black holes—represent states of much greater gravitational entropy. The early universe therefore combined an exceptionally smooth large-scale matter distribution with small fluctuations capable of growing into structure. The subsequent formation of gravitational structure is part of the physical history through which the universe developed from its unusually low-entropy initial condition into the complex, high-entropy states found later in the Block.

The primordial density fluctuations should not, however, be identified as the fundamental source of the direction of experienced time. Rather, they are an important physical ingredient in the mechanism by which the thermodynamic asymmetry of the early universe becomes expressed through the formation of structure and irreversible processes. The low-entropy boundary condition supplies the fundamental asymmetry, while the initial fluctuations allow gravitational evolution to generate the structured physical systems in which that asymmetry can be manifested.

The important point is that the thermodynamic asymmetry should not simply be identified with the direction of experienced time. Saying that entropy increases toward one end of the block does not, by itself, explain why that direction should be experienced by an observer as the future rather than the past.

Instead, the thermodynamic and gravitational asymmetries provide the physical conditions under which temporally asymmetric processes and records can arise. As structure forms and physical systems interact, information about preceding states becomes encoded in subsequent states through processes such as measurement, interaction, memory formation, and the formation of physical records. These processes are thermodynamically irreversible at the macroscopic level and therefore contribute to the asymmetric distribution of physical information across successive states.

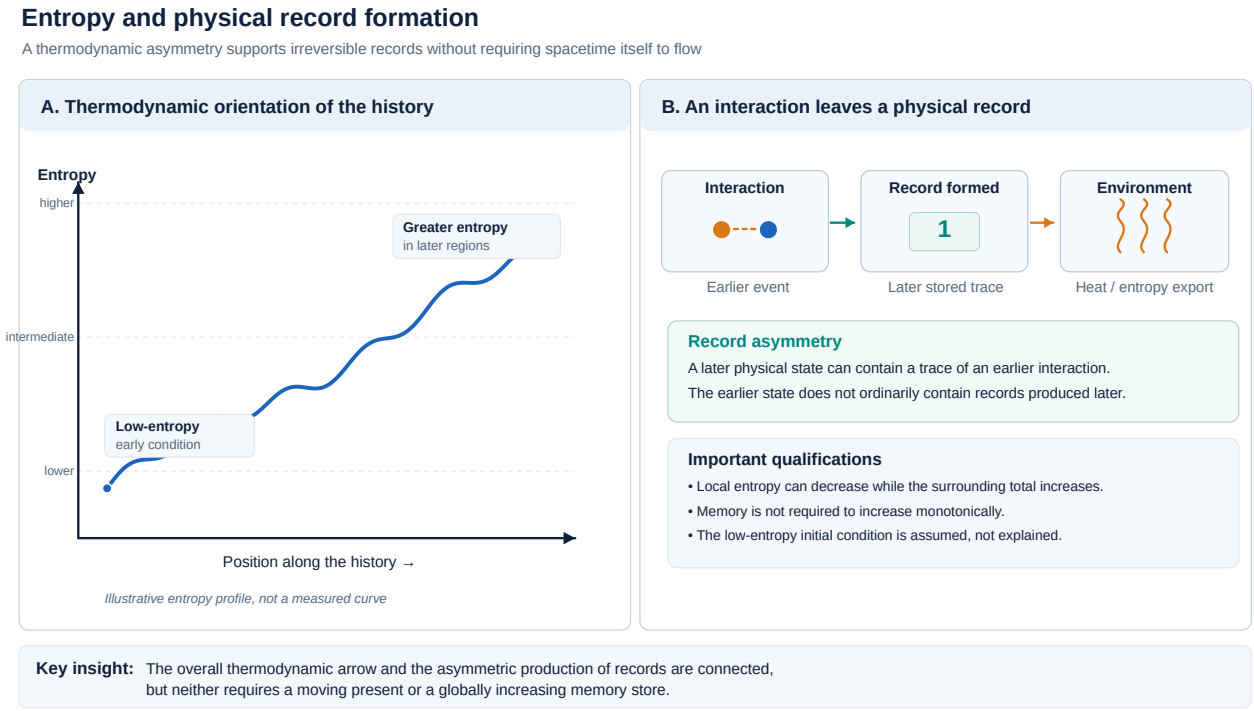


Figure 2.2. *Entropy and the formation of physical records.*
(A) An illustrative overall entropy gradient within one complete physical history; local fluctuations and exceptions are possible. (B) An interaction can create a stored physical trace while exporting entropy to the environment. Later states ordinarily contain records of earlier interactions; the Block Universe does not explain the low-entropy boundary condition itself.

Thus, the relevant chain of explanation is not simply:

Low entropy → flow of time.

Nor is it necessary to treat primordial inhomogeneity as an independent arrow of time. In the particular cosmological history of our universe, an important explanatory pathway is:

low-entropy initial condition → primordial density fluctuations → gravitational structure formation → thermodynamically irreversible processes → asymmetric formation of physical records → information asymmetry in observers → experienced direction of time.

In this formulation, entropy does not define the direction of experienced time. Rather, the low-entropy initial condition establishes the fundamental thermodynamic asymmetry of the universe, while primordial fluctuations provide the seeds through which gravitational structure and increasingly complex physical systems develop. The resulting thermodynamically asymmetric processes produce information-bearing records in a preferred temporal orientation. Human observers, as physical information-processing systems embedded within this history, inherit that information asymmetry, which in turn provides the physical basis for their experience of a directed **flow of time**.

2.6 Why Local Entropy Decreases Do Not Reverse the Flow of Time

This distinction also resolves an apparent difficulty with identifying the direction of time directly with entropy.

Entropy is not required to increase monotonically in every local region of spacetime. Local entropy can decrease, provided that this decrease is compensated by a sufficient increase elsewhere so that the total entropy of the appropriate closed system increases. Consequently, there can be local fluctuations and processes in which entropy decreases even while the overall thermodynamic arrow remains unchanged.

If the direction of experienced time were simply defined as the direction of increasing local entropy, an observer might appear to experience a reversal of time whenever it entered a region in which local entropy happened to decrease. This is not what we observe.

Our experience of time therefore cannot be identified directly with the local entropy gradient.

Instead, the relevant connection is between the thermodynamic asymmetry of the physical environment and the asymmetric formation of records within physical systems. A local fluctuation in entropy does not, by itself, reverse the causal and informational structure of an observer's physical states. The observer continues to contain records of what we call the past and not of what we call the future.

Thus, the direction of experienced time is a property of the observer's information-bearing physical state, while the thermodynamic arrow provides the broader physical asymmetry from which this information structure emerges.

2.7 Past, Present, and Future as Relations Within the Block

We can now return to the central premise of the Block Universe.

At the level of the complete four-dimensional spacetime, S_1, S_2, and S_3 all exist. Nothing has been created when the brain reaches S_2, and nothing ceases to exist when the brain passes beyond it. The entire sequence is already part of the four-dimensional structure.

Nevertheless, the brain at S_2 has a particular informational relationship to the rest of spacetime. It contains records of states associated with S_1, while it does not yet contain corresponding records of S_3. It is this asymmetric informational relationship that gives rise to the observer's distinction between past and future.

The familiar human experience of temporal passage can therefore be understood without requiring time itself to flow through the Block Universe.

The past is not a region of spacetime that has ceased to exist. The future is not a region of spacetime that has yet to come into existence. Both are already present in the block. What changes from one point along the observer's worldline to another is the physical state of the observer and the information encoded within that state.

> **Conclusion:** The Block Universe is static with respect to its complete four-dimensional existence, but it is not temporally symmetric in its physical contents. The low-entropy condition of the early universe establishes a thermodynamic asymmetry within the block. That asymmetry gives rise to a corresponding asymmetry in the formation of physical records. Human brains, as physical information-processing systems embedded within spacetime, inherit this asymmetry: later brain states contain records of earlier states, while earlier states do not contain records of later states. The

> brain consequently experiences the direction of this information asymmetry as the flow of time, distinguishing past, present, and future. **Thus, the flow of time is not a fundamental motion of the Block Universe itself, but an emergent feature of observers physically embedded within its asymmetric structure**.

3. The Laws of Physics in a Static Block Universe

3.1 How to Think About the Laws of Physics in a Static Block Universe

Because human observers perceive time as flowing, we naturally tend to think of the laws of physics in dynamic terms. We imagine the universe as having a particular physical state at the present moment, with the laws of physics determining what happens next. In this picture, the current “now” can be thought of conceptually as an input, the next “now” as an output, and the laws of physics as the rules that relate the two:

Current state → laws of physics → next state

This is a useful conceptual model of physical evolution. It corresponds naturally to our experience of time: one state exists “now,” physical processes occur, and another state subsequently becomes the new “now.”, similar to an algorithm with inputs and outputs.

This algorithm analogy is intended only to make this conceptual distinction clear. It should not be interpreted literally. The laws of physics need not constitute an algorithm in the computational sense, nor do they necessarily imply that the universe is deterministic. An algorithm, in the conceptual sense used here, could contain a probabilistic or random component. The important point is simply that the dynamic description treats the physical state at one time as being related by the laws of physics to the physical state at another time.

The Block Universe provides another way of describing the same physical reality.

If the past, present, and future all exist as parts of a complete four-dimensional spacetime, then the universe does not need to be thought of as continually generating a new “now.” There is no physical process in which the present moment disappears and is replaced by a newly created future moment. The complete history of the universe is already represented within the four-dimensional structure.

This changes the way in which we can think about the laws of physics.

Rather than regarding the laws as rules that are continually being executed as the universe advances from one moment to the next, we can regard them as being **enshrined in the self-consistent physical relationships contained within the Block Universe itself**.

For example, instead of imagining a physical object as repeatedly being updated from one position to another,

$x(T_1) \rightarrow x(T_2) \rightarrow x(T_3)$

we can regard its complete worldline as part of the four-dimensional physical structure. The laws of physics describe the relationships between the different points along that worldline and between that worldline and the other physical systems with which it interacts.

From this perspective, the laws of physics do not have to be thought of as something external to the Block Universe that is continually acting upon it. Rather, the physical contents of the block exhibit the relationships described by those laws.

Dynamic view:
The laws relate the current state to subsequent states.

Block Universe view:
The complete physical history exhibits the relationships described by the laws.

These are not necessarily competing descriptions. They are two ways of conceptualizing the same physical relationships.

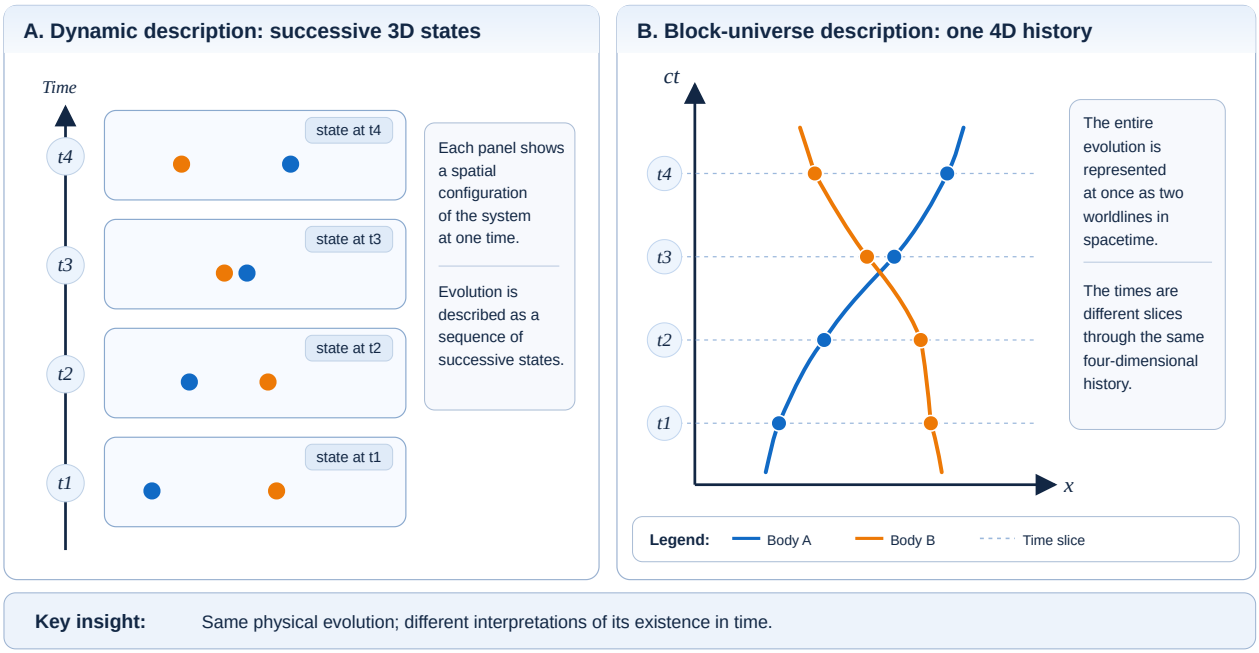


Figure 3.1. *Dynamic description and block-universe description of the same physical evolution.* (A) A system represented as a sequence of spatial states ordered by time. (B) The same system represented as a single spacetime history containing two complete worldlines. The same sampled positions appear in both panels; the difference is ontological.

The dynamic description emphasizes **evolution**. The Block Universe description emphasizes **self-consistency**.

This distinction also changes how we should think about our own experience of discovering the laws of physics. Because we experience time as flowing, we encounter the physical world one state at a time. We observe an object in one state, then observe it in another, and discover that the relationship between those states is consistent. We perform an experiment, obtain a result, repeat the experiment, and find the same physical relationship again.

In this sense, our perception of the flow of time causes us to encounter the self-consistent structure of the Block Universe sequentially. What we describe as a law of physics is, from this perspective, a relationship that remains consistently manifested throughout the relevant regions of spacetime.

We therefore do not need to imagine that the universe is repeatedly “applying” the laws of physics in real time in order for physical laws to govern what we observe. The complete spacetime structure can simply embody those relationships. As observers embedded within that structure, we encounter them sequentially and interpret their persistent consistency as the laws governing the evolution of physical systems.

This provides an important conceptual bridge between the static and dynamic descriptions of reality.

> The Block Universe is static as a four-dimensional entity, while the physical relationships within it are precisely those that, when experienced along an observer’s worldline, appear as dynamical evolution.

This distinction will become particularly important when considering quantum mechanics. If the laws of physics can be understood either as governing successive physical states or as describing the self-consistent relationships contained within the complete spacetime structure, then the apparent transition from one quantum state to another must also be examined from both perspectives. In particular, it raises the question of what, within a static four-dimensional description, the conventional notion of **wave-function collapse** is actually describing.

4. The Block Universe and Quantum Mechanics

4.1 Relativity, the Block Universe, and the Measurement Problem

The Block Universe developed in the preceding sections is based on the four-dimensional spacetime description provided by the theory of relativity. Relativity, and the Block Universe interpretation built upon it, are normally regarded as *classical* in the sense that the spacetime geometry itself is not treated as a quantum object. In particular, general relativity does not require spacetime, or the gravitational field represented by its geometry, to be quantized.

For the purposes of the argument developed in this section, however, that distinction is not central. We are not attempting to construct a deeper theory that unifies general relativity with quantum mechanics, nor are we attempting to determine the ultimate theory of quantum gravity. The fact that our present theories are expected to be incomplete at certain physical regimes does not prevent us from asking what each theory says within the domains in which it successfully applies.

Both relativity and quantum mechanics are, in important respects, understood as parts of a deeper physical description. General relativity encounters regimes in which its classical description of spacetime is expected to become inadequate, while quantum mechanics and quantum field theory do not by themselves provide a complete description of gravitation and spacetime at the most extreme scales. The existence of these limitations is an important motivation for fundamental physics, but it is not the problem we are addressing here.

Instead, the question is considerably more specific. Quantum mechanics appears to contain a peculiar transition between two apparently different kinds of evolution. According to the Schrödinger equation, a quantum state evolves continuously and deterministically according to a well-defined mathematical rule. Yet when a measurement is made, the theory is conventionally described as undergoing a sudden *collapse of the wave function* to one of the possible measurement outcomes.

This raises the familiar **measurement problem**: if the quantum state evolves according to the Schrödinger equation, and if the measuring apparatus is itself a physical system governed by quantum mechanics, what physical process causes the apparently discontinuous transition from a superposition of possible outcomes to a single observed outcome?

The problem is therefore not, at least for the purposes of this discussion, whether quantum mechanics is the final theory of nature. Nor is it necessary to solve the problem of quantum gravity. The question is whether the four-dimensional perspective developed in the preceding sections can provide a different way of understanding what is described as wave-function collapse.

In other words, we will temporarily accept the successful mathematical framework of quantum mechanics and ask whether the apparent mystery of measurement is partly a consequence of the way we ordinarily describe events as unfolding in time. The Block Universe gives us a very different starting point: rather than imagining reality as something that is continuously being created from moment to moment, we begin with the complete

four-dimensional spacetime structure. The question then becomes whether quantum measurement needs to be understood as a physical event in which the universe literally selects and creates one outcome, or whether the apparent collapse can be understood differently within the complete spacetime description.

$$\text{Unitary quantum evolution:} \quad i\hbar\, \partial|\psi\rangle/\partial t = \hat{H}|\psi\rangle$$

$$\text{Measurement:} \quad |\psi\rangle \rightarrow |\text{outcome}\rangle$$

The second transition is the feature we will examine. The aim is not to dismiss quantum mechanics, nor to replace its experimentally successful predictions with an unsupported alternative, but to investigate whether the Block Universe changes the conceptual interpretation of this transition.

4.2 Entanglement and Superposition Are Distinct Concepts

Before continuing with the discussion of the measurement problem, it is important to distinguish two concepts that are often treated as though they were interchangeable: **superposition** and **entanglement**. They are related in quantum mechanics, but they are not the same thing.

A system can be in a superposition without being entangled with another system. For example, a single quantum system may be described by

$$|\psi\rangle = (1/\sqrt{2})(|R\rangle + |B\rangle).$$

Here the quantum state is a superposition of two possible states. Nothing about this expression requires the existence of a second system.

Entanglement is instead a property of a *joint quantum state*. For a pure joint quantum state, two systems are entangled when the state cannot be expressed as a product of independent subsystem states. More generally, a mixed state is entangled if it cannot be represented as a probabilistic mixture of such product states. For example, the pure state

$$|\Psi\rangle = (1/\sqrt{2})(|R\rangle_A|B\rangle_B + |B\rangle_A|R\rangle_B)$$

cannot be decomposed into a state for A multiplied by an independent state for B. The physical description therefore belongs to the pair as a whole rather than being reducible to two independent quantum states.

This should also be distinguished from ordinary classical correlation. Two colored balls can be correlated—for example, one is red and the other blue—without either ball being in a quantum superposition and without the pair being quantum mechanically entangled. In such a case the balls possess individual physical states, and the correlation simply relates those states.

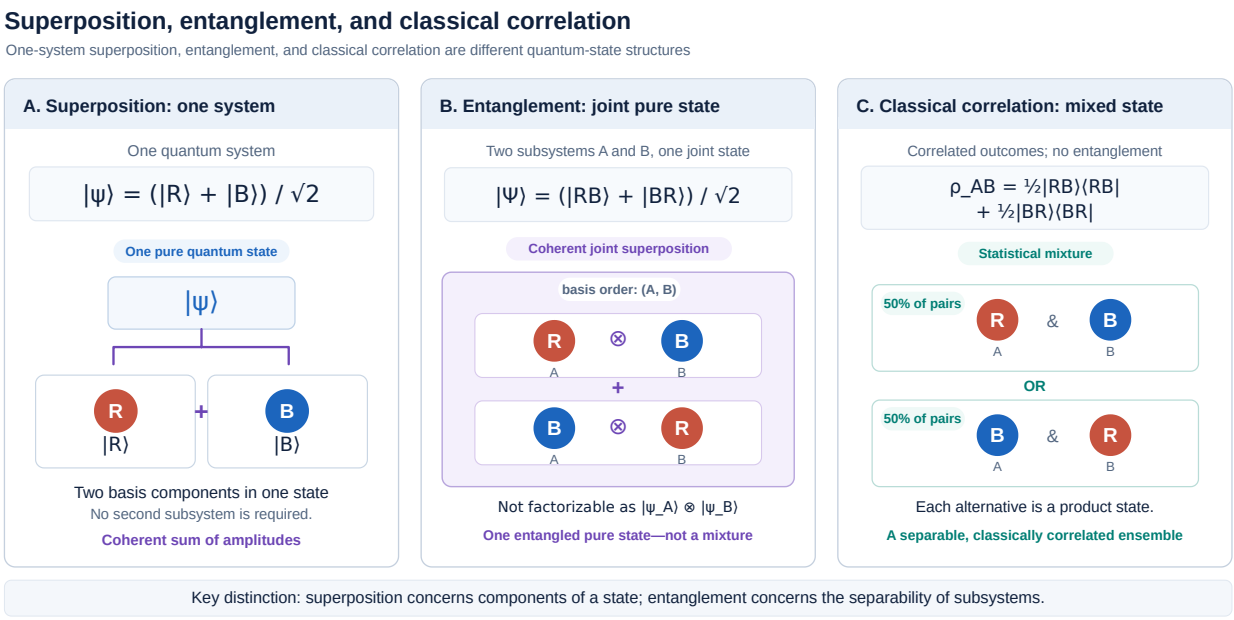


Figure 4.1. *Superposition, entanglement, and classical correlation.*

(A) A single system may be in a superposition without being entangled. (B) An entangled pure state is a coherent, non-factorizable joint state; its two terms are not preassigned classical pair types. (C) A classically correlated mixed state can instead be a probabilistic mixture of product states, and is therefore separable. The pair basis order is (A, B). R and B denote the abstract basis labels used in Section 4.2; colored symbols are visual aids, not particle trajectories.

Correlation ≠ Superposition Entanglement ≠ Superposition

Quantum entanglement can involve superposition, but the defining feature of entanglement is the non-separability of the joint quantum state. This distinction becomes important when interpreting Bell experiments. Bell's theorem is not simply a statement that quantum systems are in superposition. It concerns whether the observed correlations of entangled systems can be reproduced by a suitable class of underlying local hidden-variable descriptions.

4.3 Bell's Theorem and the Assumptions Behind Bell Inequalities

Bell's theorem addresses a precise question: can the correlations predicted by quantum mechanics, and observed experimentally, be reproduced by a theory in which measurement outcomes arise from underlying physical variables while respecting locality and certain statistical assumptions?

Let λ denote the relevant underlying physical state of a pair of systems. Alice chooses a measurement setting a and obtains an outcome A; Bob chooses a measurement setting b and obtains an outcome B. In a conventional local hidden-variable description, the outcomes are determined probabilistically or deterministically by the local setting and the underlying state. In the deterministic special case this may be written schematically as

$$A = A(a, \lambda), \quad B = B(b, \lambda).$$

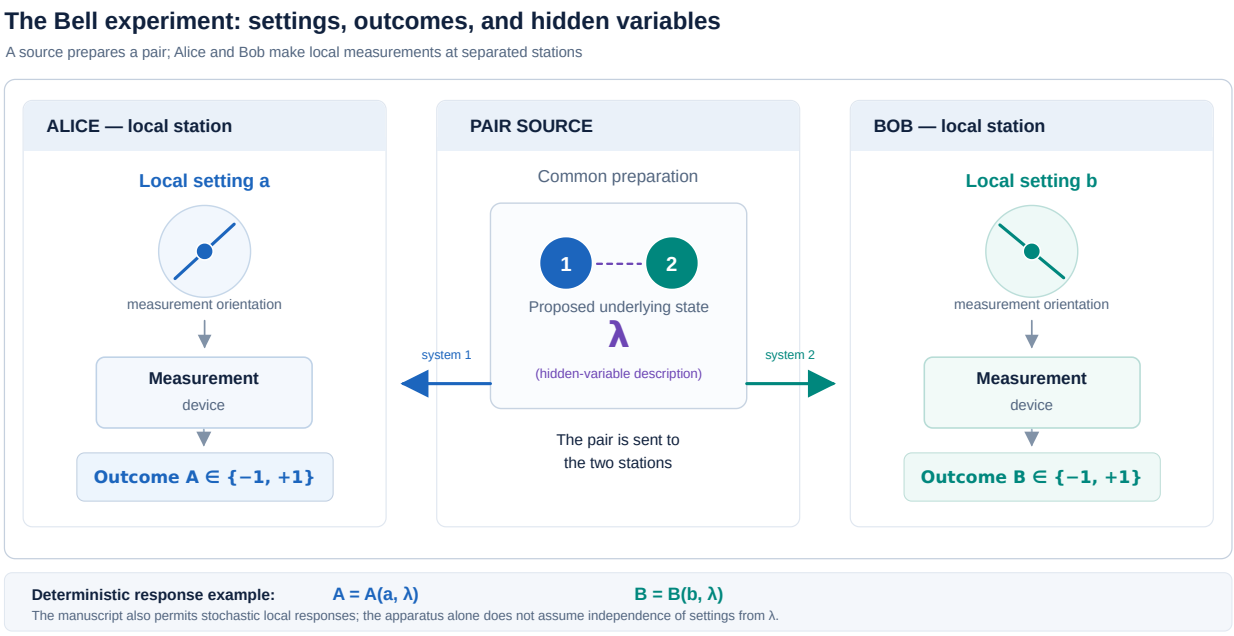


Figure 4.2. *Bell-test arrangement, measurement settings, outcomes, and proposed hidden variables.*

A source prepares a pair sent to spatially separated stations. Alice selects setting a and records outcome A; Bob selects setting b and records outcome B. In a local hidden-variable description, λ denotes the relevant underlying state of the pair, and local responses use (a, λ) and (b, λ). The diagram does not assume measurement independence P(λ|a,b) = P(λ), nor does it depict communication between the stations.

The locality condition relevant to Bell's theorem is more precise than the ordinary statement that faster-than-light communication is impossible. Bell locality requires that, once the relevant underlying physical state λ is specified, the joint probability of the outcomes factorizes into independent local response probabilities:

$$P(A,B \mid a,b,\lambda) = P(A \mid a,\lambda)\, P(B \mid b,\lambda).$$

In this formulation, Alice's local outcome probabilities depend on her setting and λ, while Bob's depend on his setting and λ. The distant measurement setting or outcome does not provide additional information about the local outcome once these variables have been specified.

This condition should be distinguished from operational no-signaling, which requires that the observable probability distribution of Alice's outcomes cannot be changed by Bob's choice of measurement setting, and vice versa. No-signaling is a weaker condition than Bell locality. Quantum mechanics respects operational no-signaling even though its entangled-state correlations cannot be reproduced by Bell-local, measurement-independent hidden-variable models.

This distinction is important for the present argument. Our objective is not merely to preserve the impossibility of faster-than-light communication, but to examine whether a local hidden-variable description can reproduce quantum correlations when measurement independence is relaxed.

Bell inequalities require more than Bell locality alone. A central additional assumption is measurement independence, also called statistical independence:

$$P(\lambda \mid a,b) = P(\lambda).$$

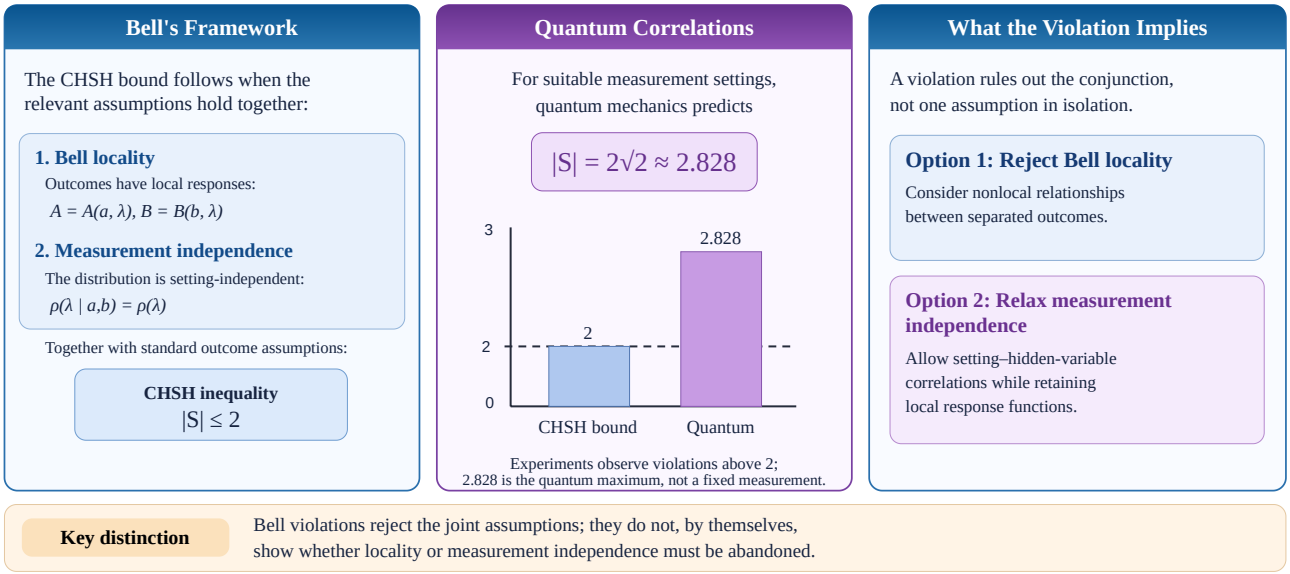
This expresses the assumption that the underlying physical state λ is statistically independent of the measurement settings selected by Alice and Bob. It is often described informally in terms of “free choice,” but the mathematical content is statistical rather than metaphysical. The assumption is not a consequence of special relativity.

Bell's original 1964 argument treated some of these assumptions implicitly, while later formulations made the statistical assumptions more explicit. For present purposes, the important logical point is that the Bell inequality follows only for the class of theories satisfying the relevant Bell assumptions. Consequently, an experimental violation establishes that the observed correlations cannot be reproduced by a theory satisfying all of those assumptions simultaneously.

Bell inequality violated ⇒ at least one required assumption fails.

This statement is weaker, and more precise, than the frequently encountered claim that Bell's theorem proves that physical influences must travel faster than light. Bell-inequality violations establish correlations that cannot be reproduced by Bell-local, measurement-independent hidden-variable models. They do not by themselves identify which assumption must be abandoned, nor do they establish that a usable faster-than-light signal has been transmitted.

Figure 4.3: Why Bell Inequalities Can Be Violated



4.4 The Experimental Result: What the Bell Correlations Actually Tell Us

The experimental result should first be taken at face value. Quantum experiments produce correlations between spatially separated measurement outcomes that violate Bell inequalities. Whatever interpretation is ultimately adopted, the observed correlations are an empirical fact that any adequate physical account must reproduce.

The usual conceptual question is then framed as follows: if Alice and Bob are separated by a spacelike interval, how can their outcomes be correlated more strongly than permitted by a local hidden-variable model?

A common response is to attribute the correlations to a nonlocal feature of the quantum state, sometimes described informally as “spooky action at a distance.” But the experimental result itself is more limited. It tells us that the correlations cannot be explained by Bell-local hidden-variable models satisfying measurement independence. It does not by itself establish that one measurement physically changes the distant system instantaneously.

This distinction motivates the alternative developed here. Rather than beginning by asking what physical influence travels from Alice to Bob, we ask whether the observed correlation could instead be understood as a relationship among events already contained within a complete four-dimensional physical history.

4.5 The Block Universe and Measurement Independence

The Block Universe provides a natural perspective from which to question measurement independence as a fundamental requirement. In the four-dimensional description developed earlier, the complete physical history contains events at different spacetime locations and times. The fact that one event occurs later along a particular worldline does not mean that it must be statistically independent of every earlier physical variable.

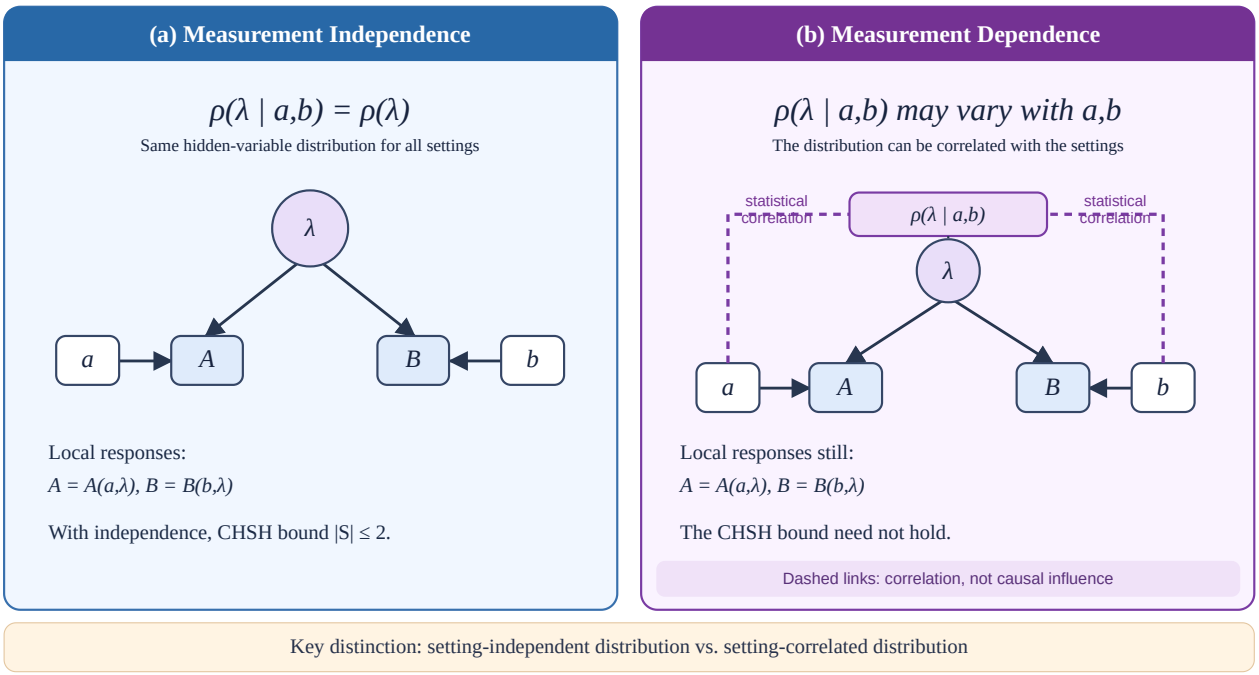
Suppose λ describes an underlying physical property of a quantum system and a and b describe measurement settings that occur later in the experimental history. The conventional statistical-independence assumption requires

$$P(\lambda \mid a,b) = P(\lambda).$$

The Block Universe proposal considered here instead allows

$$P(\lambda \mid a,b) \neq P(\lambda).$$

Figure 4.4: Measurement Independence vs. Measurement Dependence



It is important to distinguish what the Block Universe permits from what it logically requires. The existence of a complete four-dimensional physical history does not, by itself, imply that measurement independence must be violated. A Block Universe can contain statistically independent variables just as it can contain correlated ones. The proposal developed here is therefore not that measurement dependence follows necessarily from the Block Universe, but that the Block Universe provides an ontological framework in which such dependence can be considered without automatically invoking a physical influence propagating between distant events.

The meaning of this is important. It does not mean that Alice's or Bob's later measurement sends a physical influence backward in time to alter λ. Nor does it require us to postulate a new force or signal. Rather, the proposal is that the underlying physical variables and the measurement settings may possess statistical correlations as part of the complete four-dimensional history.

One possible physical origin of such correlations is shared causal ancestry. The particles, their source, the measurement apparatus, and the physical systems used to select measurement settings are not entirely isolated entities. Their past light cones may overlap, and their physical histories may include common causes and earlier interactions. Such shared causal histories can, in principle, establish statistical correlations that persist into the experiment. From the Block Universe perspective, these correlations would be ordinary relationships among physical events, not evidence of advance knowledge, backward causation, or a conspiracy between particles and experimenters.

This possibility is offered only as an illustration, not as an established explanation of Bell inequality violations. Shared causal ancestry does not, by itself, demonstrate the particular measurement dependence needed to reproduce quantum correlations. Establishing that connection would require a developed physical model, which is beyond the scope of this manuscript. The present argument requires only that measurement dependence remains a physically conceivable alternative to abandoning locality, not that its precise mechanism has been identified.

This is the central conceptual move of the present argument:

Correlation across spacetime ≠ interaction across spacetime.

In an ordinary dynamical description, one naturally asks what event causes another event to occur. In a Block Universe description, relationships between events can instead be considered as properties of the complete spacetime structure. The absence of a dynamical influence from A to B therefore does not imply the absence of a physical correlation between A and B.

However, this ontological possibility should not be confused with a complete physical explanation. Allowing measurement dependence does not explain why the correlations take precisely the form predicted by quantum mechanics. A fully developed physical theory would need to specify the relevant probability distributions and reproduce the experimentally observed correlations, including their no-signaling properties.

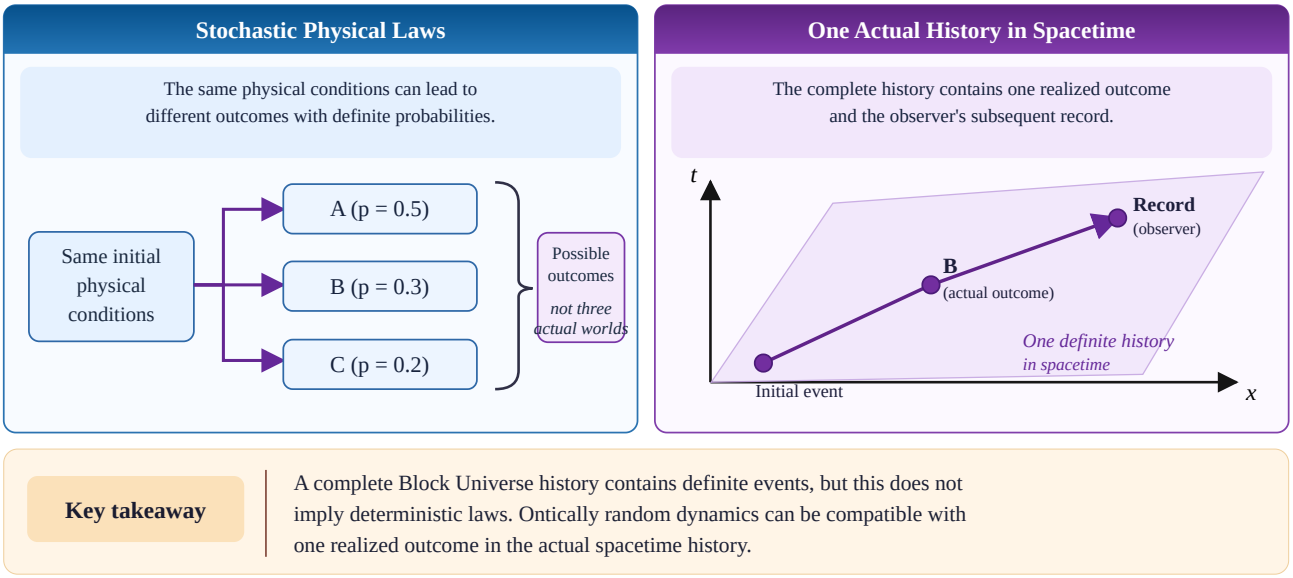
The present proposal makes a narrower claim. It questions whether measurement independence must be regarded as a fundamental physical principle and explores the consequences of relaxing it within a globally consistent Block Universe. Once measurement independence is relaxed, the conventional Bell inequality no longer constrains every local hidden-variable description in the same way.

The Block Universe thus provides a possible ontological interpretation of measurement-dependent correlations, without claiming that the Block Universe alone derives those correlations or establishes that measurement independence is actually violated in nature.

4.6 Ontically Random Hidden Variables in the Block Universe

The hidden variables considered here are not intended to restore a deterministic theory beneath quantum mechanics. This distinction is essential. The proposal retains the possibility that quantum randomness is **ontic**: the underlying physical variable itself can be genuinely random rather than merely unknown because of incomplete information.

Figure 4.5: Ontically Random Hidden Variables in the Block Universe



Key takeaway

A complete Block Universe history contains definite events, but this does not imply deterministic laws. Ontically random dynamics can be compatible with one realized outcome in the actual spacetime history.

Consider a simplified double-slit experiment involving many photons. Introduce an underlying variable λ with a hundred possible values, where each value labels a possible complete experimental history, including the measurement arrangement and the accumulated detection records. Values 1–10 might correspond to histories in which the experimental record indicates the photon went through the left slit; values 11–20 might correspond to histories indicating the right slit; and values 21–100 might correspond to histories in which no which-slit measurement is performed and the accumulated detections of many photons produce an interference pattern. The last category makes no claim about a particular classical trajectory through both slits. It is merely a label for the observed interference outcome.

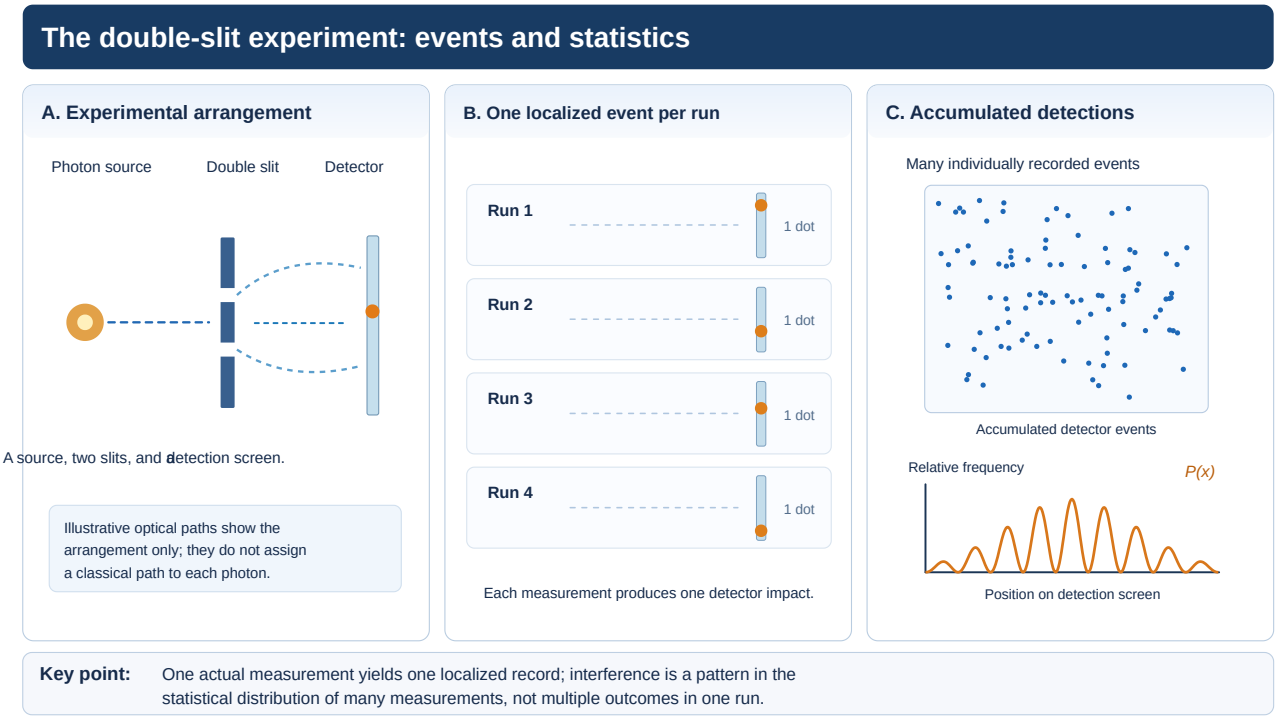


Figure 4.6. *Individual detection events and the double-slit interference pattern.* (A) The arrangement contains a source, two slits, and a screen. (B) A trial produces one localized detection. (C) Many trials yield the familiar interference distribution P(x). The actual history contains definite records; quantum mechanics predicts their statistics.

The particular value of λ is taken to be genuinely random. Before the relevant event is encountered along an observer's worldline, the observer cannot know which value will be realized. The point is not that the value was secretly determined by some deeper deterministic variable.

What is added is a correlation between λ and the measurement history. If the experimental arrangement includes a measurement of which slit the photon traverses, the value of λ encountered by the observer is correlated with that measurement arrangement. When the measurement records the left slit, the observer discovers a value in the corresponding group. The measurement does not cause the photon to decide, at that instant, which history it had followed. Rather, the measurement event and the photon history are components of one self-consistent Block Universe history.

The same conceptual structure can be extended to Bell experiments. Let λ be an ontically random hidden variable associated with an entangled pair, while a and b are Alice's and Bob's measurement settings. The complete history may contain correlations among

$$\lambda, \quad a, \quad b, \quad A, \quad B.$$

Those correlations do not require a signal to propagate from Alice to Bob, nor from Bob to Alice. They are simply correlations among physical events contained in the same four-dimensional history. Quantum mechanics supplies the experimentally successful probability structure that those correlations must reproduce, while the hidden variable identifies which particular physical history is encountered along an observer's worldline.

In this picture, quantum mechanics need not be replaced by a deeper deterministic theory. The proposal instead retains the quantum probability distribution and adds an ontological interpretation of the individual outcomes. The hidden variable answers the question, “Which particular outcome occurs in this physical history?” Quantum mechanics answers the statistical question, “With what probabilities are the different outcomes realized?”

$$\text{Quantum mechanics} \rightarrow \text{statistical structure of possible histories}$$

$$\text{Block Universe + hidden variables} \rightarrow \text{particular realized history}$$

This is therefore distinct from a proposal in which quantum probabilities are merely epistemic and a deeper deterministic theory is assumed to exist. Such a theory might, in principle, resemble the relationship between microscopic dynamics and macroscopic statistical laws, or between detailed physical dynamics and probabilistic weather forecasts. That is a much stronger hypothesis and is not required for the present argument.

4.7 Measurement as a State Transition Along a Worldline

The Block Universe perspective also changes the conceptual status of measurement itself. In the ordinary dynamical picture, a measurement is often represented as a sudden transition of the quantum state:

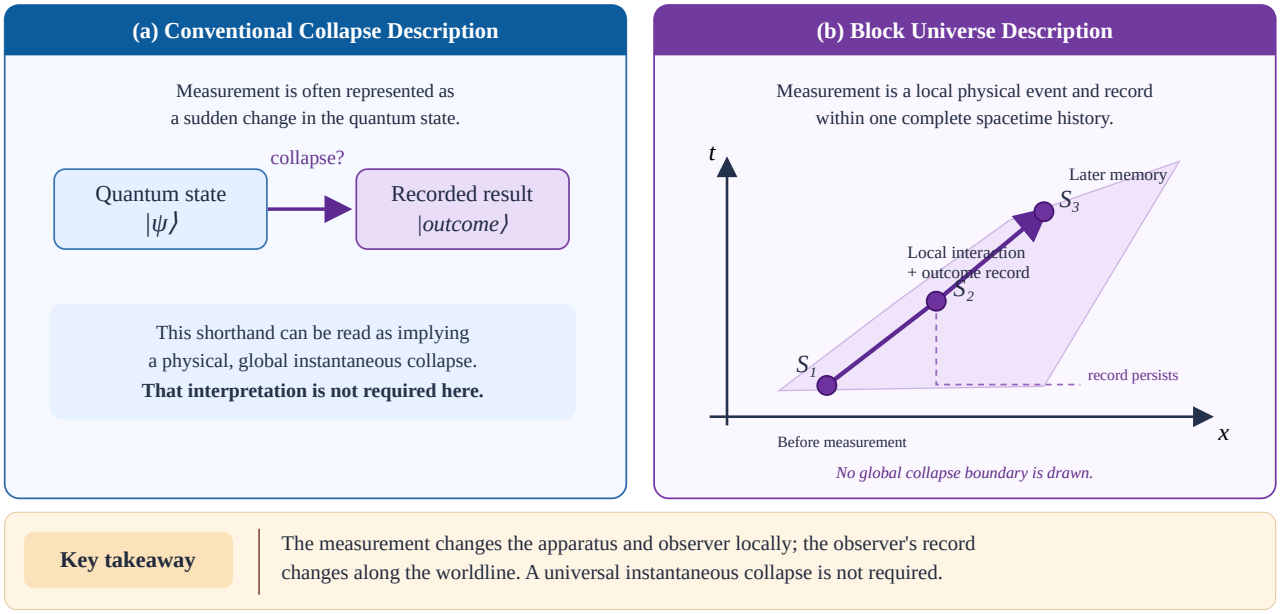
$$|\psi\rangle \rightarrow |\text{outcome}\rangle.$$

This notation naturally suggests that the universe was previously in a superposition and that, at the instant of measurement, some physical process caused one possibility to become real while the others ceased to be real. This is the source of much of the mystery associated with wave-function collapse.

In the Block Universe picture developed here, the measurement can instead be understood as a physical correlation between the measured system, the measuring apparatus, and the observer. The observer is itself a physical system with a worldline. Successive states of that observer may be represented schematically as

$$S_1 \rightarrow S_2 \rightarrow S_3.$$

Figure 4.7: Measurement Along an Observer's Worldline



Key takeaway The measurement changes the apparatus and observer locally; the observer's record changes along the worldline. A universal instantaneous collapse is not required.

At S_2, the observer has a physical record of a measurement outcome. The observer therefore experiences the measurement as an event occurring at a particular time. But from the four-dimensional perspective, there is no requirement that the entire universe undergo a sudden, globally propagating collapse at S_2. The measurement is one part of the complete spacetime structure.

This does not deny that a quantum measurement produces a physical change in the measuring apparatus or observer. It denies only that this local physical change must be accompanied by an instantaneous change propagating through the entire universal quantum state.

The proposal can therefore be summarized as follows:

Quantum state and its evolution → statistical constraints on physical histories

Ontically random hidden variable → identification of the particular outcome in the actual history

Block Universe → complete four-dimensional consistency of the measurement history

Observer's worldline → sequential encounter with the physical states and records constituting the measurement

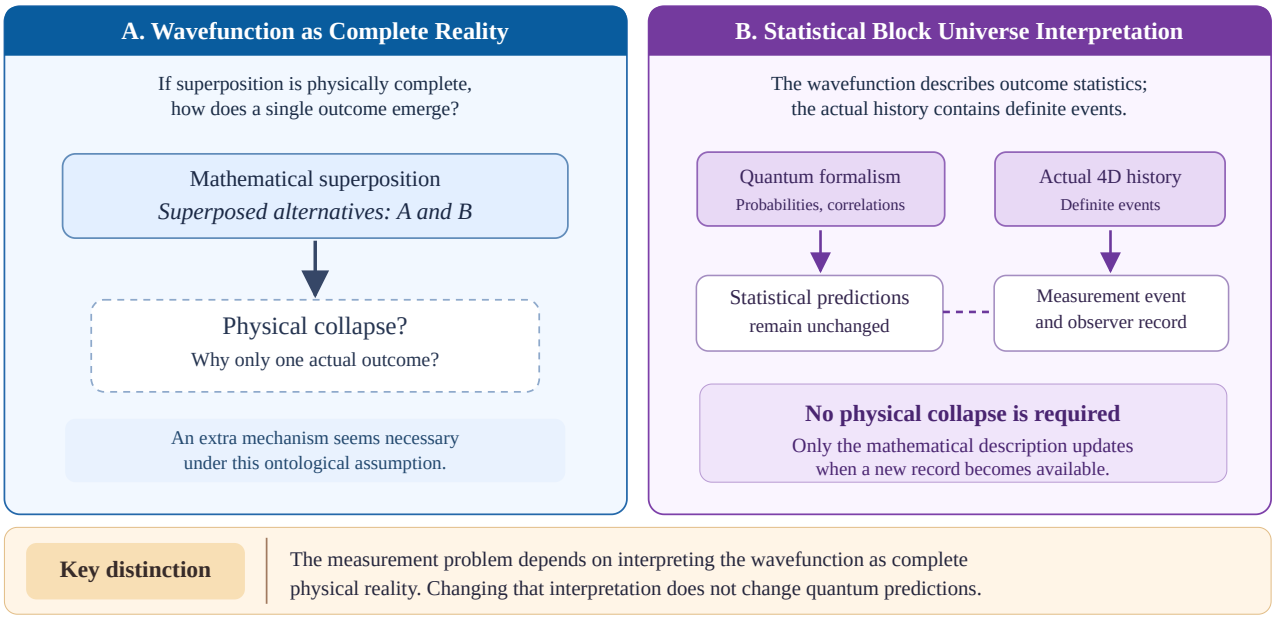
On this interpretation, wave-function collapse is not a fundamental physical process occurring throughout spacetime. What is conventionally described as collapse is instead a mathematical representation of the change in information available to the observer as a result of the physical state transition along the observer's worldline.

4.8 Dissolving the Measurement Problem

The traditional quantum measurement problem arises from the apparent conflict between the continuous, unitary evolution of the wavefunction and the definite outcomes recorded in physical measurements. This conflict arises when the wavefunction is assumed to provide a complete description of physical reality, so that a mathematical superposition must somehow become a single actual physical outcome.

The Block Universe interpretation developed here rejects that assumption. The wavefunction is understood as a mathematical description of possible measurement outcomes and their statistical relationships, rather than as a physical entity whose superposition must collapse. The actual four-dimensional universe contains one complete physical history, including definite measurement events, interactions, and physical records. These outcomes do not require a separate physical process to bring them into existence.

Figure 4.8: Dissolving the Measurement Problem



Quantum mechanics continues to describe the statistical structure of these outcomes using its existing mathematical formalism. The Schrödinger equation, Born rule, quantum interference, and experimentally observed correlations remain unchanged. What changes is the ontological interpretation of the mathematics, not the mathematics itself.

Within this framework, the traditional measurement problem is therefore dissolved rather than solved by introducing a new physical mechanism. There is no need to explain how a physically existing superposition becomes a definite outcome, because the superposition is not assumed to represent multiple simultaneously realized physical outcomes. The measurement itself remains a real physical interaction, while wavefunction collapse represents an update of the mathematical description associated with the observer's acquisition of a definite physical record.

This conclusion does not establish that the Block Universe interpretation is uniquely correct, nor does it rule out alternative interpretations of quantum mechanics. It establishes a narrower but significant point: **once the wavefunction is no longer regarded as a complete description of physical reality, the traditional demand**

for a physical collapse mechanism disappears.

4.9 The Scope and Test of the Proposal

The argument developed so far should be stated modestly. It does not claim that Bell's experiments are mistaken, that quantum mechanics is empirically incorrect, or that a complete alternative theory has already been constructed. The experimental correlations remain exactly what they are: correlations that violate Bell inequalities.

The proposal instead questions the interpretation of those violations. Bell's result shows that the observed correlations cannot be reproduced by hidden-variable theories satisfying all of the assumptions required to derive Bell's inequality. The present proposal retains locality and quantum randomness but rejects measurement independence, allowing the hidden variables to be statistically correlated with the measurement settings as part of the complete Block Universe history.

The Block Universe interpretation:

- The actual physical universe consists of one complete four-dimensional history containing definite measurement events and outcomes.
- The wavefunction describes the statistical structure of possible outcomes rather than constituting a complete description of physical reality.
- Wavefunction collapse is a mathematical update, not an additional physical process that creates a definite outcome.
- The established quantum mathematical formalism and its experimental predictions remain unchanged.

The additional proposal examined in the Bell analysis:

- Hidden variables identify the particular outcomes belonging to the actual physical history.
- These variables may be ontically random.
- Measurement independence is relaxed, allowing statistical dependence between hidden variables and measurement settings.
- Local response functions are retained, without introducing a superluminal signaling mechanism.
- The Appendix A construction demonstrates that this approach can reproduce the quantum singlet correlations.

The second proposal is not a logical consequence of the first. The Block Universe does not require measurement dependence, nor does the interpretation of wavefunction collapse as a mathematical update depend on establishing measurement dependence. Rather, the Block Universe provides an ontological setting in which measurement-dependent correlations can be considered without requiring instantaneous physical interactions between distant measurement events.

The crucial mathematical test is whether this framework can reproduce the precise correlations observed in Bell experiments while maintaining the stated locality condition. The CHSH formulation provides the cleanest way to make that test explicit. The derivation of the CHSH bound, the exact role of measurement

independence, and an explicit measurement-dependent local hidden-variable construction are presented in Appendix A. The main text will use those results rather than interrupt the conceptual development with the full algebra.

4.10 Delayed-choice and Quantum Eraser Experiments

In these experiments, quantum systems are prepared so that which-path information is correlated with another degree of freedom or an entangled partner. A subsequent measurement can be arranged so that the resulting data distinguish paths or reveal interference within appropriately selected subsets.

In some arrangements, the relevant measurement occurs after the signal particle has already been detected. This can create the impression that the later measurement somehow changes whether the earlier particle exhibited wave-like or particle-like behavior. However, the experimentally relevant result is a correlation between measurement records. The unconditioned distribution of earlier detection events does not change according to the later measurement choice. Interference is revealed in particular subsets of the data when the corresponding measurement records are compared.

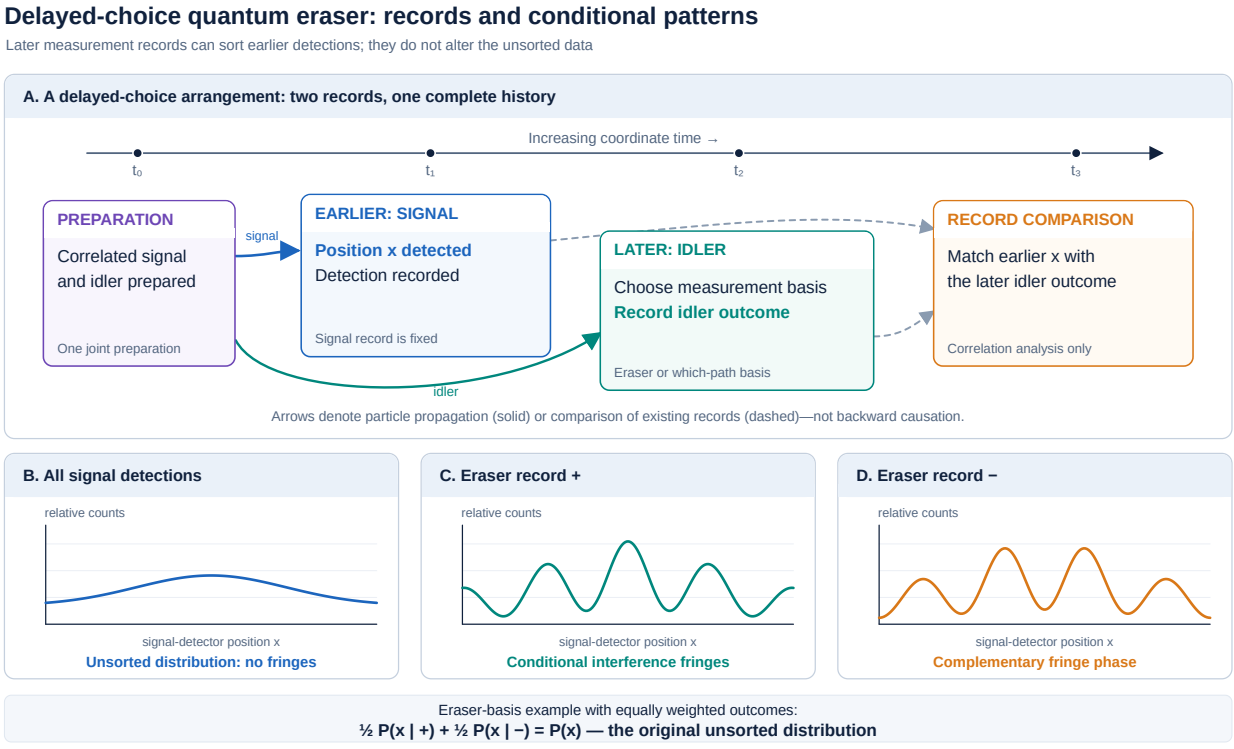


Figure 4.9. *Delayed-choice quantum eraser: records and conditional detection patterns.*
In a delayed-choice arrangement, the signal is recorded before the idler measurement. Sorting the fixed earlier detections by later idler records can reveal complementary interference patterns in selected subsets. Their weighted sum reproduces the unchanged unsorted distribution; no earlier event is modified.

The Block Universe provides a natural interpretation. All the detection events, measurement settings, and correlations are already part of the complete four-dimensional history. Comparing the records reveals relationships among those events; it does not create or modify the events themselves. No physical signal needs to travel backward through time, and no earlier measurement record needs to be rewritten.

The Block Universe framework developed in this section also clarifies the role of wave-function collapse in these experiments. Quantum mechanics uses the wave function to describe the statistical structure of possible measurement outcomes. When an outcome is obtained, the mathematical formalism updates the quantum state accordingly. This update need not be interpreted as a physical process propagating instantaneously through spacetime. In the proposed Block Universe account, the actual physical history already contains the particular measurement settings, interactions, and outcomes that occurred. The mathematical collapse reflects conditioning the quantum description on an observed result, rather than a physical operation that creates the corresponding history. The delayed-choice aspect therefore introduces no additional physical mystery requiring a special collapse mechanism.

The same conceptual framework applies to the Bell experiments discussed earlier. In both cases, the actual measurement settings and outcomes are components of one complete spacetime history. Our proposal allows correlations between hidden variables and measurement settings, rather than assuming measurement independence as a fundamental requirement.

The delayed-choice and quantum eraser experiments do not independently establish that measurement independence is violated. Their observed correlations can be calculated using standard quantum mechanics. Their significance for the present argument is conceptual: they illustrate how apparently paradoxical relationships between temporally separated events can be understood without requiring physical influences to propagate backward in time.

The Block Universe does not change the experimental predictions. It changes the proposed ontology of the events to which those predictions refer.

Delayed-choice and quantum eraser experiments appear mysterious when physical reality is imagined as a sequence of states that must be progressively created, with later measurements somehow determining the status of earlier events. Within the Block Universe, this difficulty is substantially reduced. The complete physical history contains the actual measurement settings and outcomes. Their correlations are properties of that history, while the quantum formalism describes their statistical relationships. There is no need to introduce a physical mechanism by which a later choice alters an earlier event, or by which mathematical wave-function collapse propagates through spacetime. The apparent paradox arises not from the observed correlations themselves, but from the additional demand that those correlations must be produced by a dynamical influence between the events.

4.11 The Core Result of Section 4

We can now state the principal result of the argument developed in this section. The detailed CHSH derivation and the explicit construction are given in Appendix A; here we summarize what that analysis establishes.

Bell's theorem does not establish that quantum correlations are impossible for every local hidden-variable theory. It establishes that they are impossible for the narrower class of local hidden-variable theories that also satisfy measurement independence and the other assumptions required by the Bell inequality.

Once measurement independence is relaxed, an explicit local-response model can reproduce the exact singlet probabilities:

$$P(A=a,B=b|x,y) = [1 - ab(x\cdot y)]/4.$$

The model does this through a setting-dependent hidden-variable distribution:

$$\rho(\lambda|x,y) = [1 - \lambda_1\lambda_2(x\cdot y)]/4.$$

The local response functions remain

$$A=\lambda_1, \quad B=\lambda_2.$$

Therefore, the observed Bell correlation need not be represented mathematically as a direct dependence of Alice's outcome on Bob's setting or vice versa.

4.12 What Has Been Shown—and What Has Not

It is useful to separate the conclusions that follow from the mathematics from the larger interpretation being proposed. The technical derivation supporting these conclusions is collected in Appendix A.

Shown by the construction:

· Measurement-independent local hidden-variable models satisfy the CHSH bound.

· Quantum mechanics predicts, and experiments observe, violations of that bound.

· The CHSH derivation uses a common hidden-variable distribution across the different setting combinations.

· Relaxing measurement independence removes that particular step.

· An explicit measurement-dependent model with local response functions reproduces the singlet joint probabilities exactly.

· The model preserves the usual local marginal statistics and therefore does not create an operational signaling channel.

What Would Be Required for a Complete Physical Theory

The construction presented in Appendix A establishes that the quantum singlet correlations can be reproduced using local response functions when measurement independence is relaxed. This demonstrates the mathematical possibility of the proposed approach, but it does not constitute a complete underlying physical theory.

To develop the proposed hidden-variable account into such a theory, the following would be required:

· **A physical specification of the hidden variables and their correlations with measurement settings.** The theory would need to identify the relevant physical variables and provide a physically grounded account of the statistical relationships represented by the setting-dependent distribution. The shared causal ancestry discussed in Section 4.5 illustrates one possible direction, but it is not yet a developed physical model.

· **A general mathematical formulation.** The theory would need to reproduce the experimentally established predictions of quantum mechanics beyond the particular singlet correlations demonstrated in Appendix A, including other measurement arrangements and physical systems.

· **Consistency with established physical constraints.** The theory would need to specify how its physical variables and correlations operate consistently with relativistic causal structure, while preserving the experimentally observed absence of superluminal signaling.

These requirements concern the development of a complete underlying physical theory, not the validity of the Block Universe as an ontological interpretation. The interpretation developed in this manuscript retains the existing quantum mathematical framework and its empirical predictions. It does not require an independently constructed hidden-variable theory to interpret wavefunction collapse as a mathematical update rather than a physical process.

As argued in Section 4.8, the traditional measurement problem is dissolved within this interpretation by rejecting the assumption that the wavefunction is a complete description of physical reality. Constructing a deeper physical theory would be a separate undertaking.

The mathematics also raises a separate philosophical question: if a complete four-dimensional physical history contains the future as well as the past, in what sense are the choices made by observers genuinely free? Section 5 addresses that question.

4.13 The Hidden Variable as Part of a Complete History

The complete history can now be represented schematically, building on the technical construction developed in Appendix A:

H = {spacetime geometry, preparation, λ, settings, outcomes, records, observer states}.

From the perspective of an observer moving along a worldline, these components are encountered sequentially. From the Block Universe perspective, they form one complete four-dimensional physical structure.

The hidden variable therefore does not need to be imagined as a little object carrying a secret message from the source to the measurement devices. It is better understood, at this stage, as part of the specification of the complete physical history.

The key unresolved issue is then no longer simply how the hidden variable travels from one place to another. The deeper question becomes why the complete history has the correlations that it does.

5. The Block Universe and Free Will

What does a four-dimensional universe imply about human choice?

5.1 The question must begin with a definition of free will

The question of whether the Block Universe is compatible with free will cannot be answered without specifying what free will means. The term is used for several different claims: that people deliberate and act on reasons; that their actions can be attributed to them; that alternative outcomes may be physically possible; or, more strongly, that an agent possesses a special power to select which outcome becomes actual.

These claims should not be conflated. In particular, the existence of deliberation does not by itself establish a freedom that stands apart from the physical history. To keep the distinction visible, we will use three terms:

Deliberative will refers to the real physical processes through which a conscious observer evaluates alternatives, drawing on conscious and unconscious information processing, and arrives at a choice. The term does not imply any exemption from physical laws.

Perceived free will refers to the observer's subjective experience of having controlled which outcome occurred. The experience itself is real; whether its implied power of selection exists is a separate question.

Fundamental free will, in the strong sense examined here, refers to an additional power of an agent to determine which physically permissible outcome becomes actual, over and above the physical processes that constitute the agent and the realized history.

In discussing a specific choice between A and B, we will use decision for the point in the actual physical history at which A rather than B is established as the realized outcome. Prior intentions, reconsiderations, and apparent commitments are physical events too, but for this purpose they belong to the deliberative history leading to the outcome; they are not privileged as separate decisions. What counts as the outcome must, of course, be specified for the question under examination.

These definitions are analytical choices for the present discussion, not universal definitions used in philosophy or cognitive science. We will first review established conceptions of free will fairly, then explain the particular interpretation suggested by the physicalist Block Universe developed in this manuscript.

5.2 Physicalist free will

The first broad interpretation is that the conscious entity is entirely physical. Its deliberation is performed by physical processes, most plausibly processes occurring in the brain, and those processes obey the laws of physics.

Under this interpretation, there is no additional non-physical agent that has to intervene in the physical process in order for a choice to occur.

5.2.1 Physicalist-deterministic free will

In a deterministic physical theory, the complete physical state of the system together with the physical laws determines its subsequent physical states. A simplified representation is:

$$S(t0) + L \rightarrow S(t1)$$

For a conscious entity, those states include the physical processes associated with perception, memory, reasoning, deliberation, preferences, intentions, decisions, and actions.

There is no conflict between this picture and the Block Universe. Indeed, the Block Universe is naturally capable of representing such a deterministic history. The successive states of the entity form part of its world line:

$$S1 \rightarrow S2 \rightarrow S3 \rightarrow \cdots$$

The crucial point is that the Block Universe itself is indifferent to determinism. The Block Universe describes the complete four-dimensional history; it does not, by itself, specify whether the laws that generate or constrain that history are deterministic.

5.2.2 Physicalist-indeterministic free will

The same is true if the underlying physical laws contain an irreducibly random, or ontically random, component.

Instead of a physical state having only one possible successor, the laws might assign probabilities to several possible successors:

$$S(t0) + L \rightarrow \{S1, S1', S1'', \cdots\}$$

The Block Universe does not object to this. The actual four-dimensional universe simply contains the particular history that occurs.

This is consistent with the distinction developed in Section 4 between randomness and independence. A variable may be genuinely random and nevertheless have a definite value in the actual history.

The Block Universe describes the actual history; it does not require the laws governing that history to be deterministic.

Thus, neither deterministic physicalism nor indeterministic physicalism is in conflict with the Block Universe merely by virtue of being deterministic or indeterministic.

5.3 Compatibilist free will

Compatibilism identifies free will with meaningful forms of agency exercised through a person's own reasons, values, preferences, and deliberation, rather than with a power to escape physical causation. On this account, an action may be free even when the processes producing it are entirely physical. Freedom may also depend on the absence of coercion or manipulation. Compatibilists therefore need not claim that an agent stands outside the laws of physics.

From the perspective adopted here, this is a defensible description of a genuine capability, but calling it free will risks confusing it with the stronger notion of an independent power to select an outcome. We prefer deliberative will for the physical capacity and perceived free will for the accompanying subjective experience. This is a terminological and interpretive preference, not a refutation of compatibilism.

The brain's organization, memories, preferences, conscious evaluation, and unconscious processing are all parts of one physical system. The detailed mechanism by which this system generates a subjective sense of agency is not fully understood. Nevertheless, it is reasonable within a physicalist account to treat that experience as emerging from the system, without claiming to have derived it from neuroscience.

The analogy with Section 2 is illuminating. Asymmetric physical records along an observer's world line help explain why time is experienced as flowing, even though the Block Universe requires no objectively advancing present. Likewise, the physical history contains representations of alternatives, deliberation, and the realized choice; an observer can experience this as exercising free will without that experience establishing an additional power outside the physical history.

$$\text{experienced flow of time} \leftrightarrow \text{asymmetric physical records}$$

$$\text{perceived free will} \leftrightarrow \text{physical records of deliberation and realized choice}$$

The analogy does not assert that the two experiences have identical neural mechanisms. Its point is ontological: a real subjective experience need not imply that the additional fundamental property attributed to it exists.

5.4 Libertarian free will

A stronger conception of free will holds that a person is genuinely capable of making alternative choices. In its strongest form, libertarian free will requires that the person be an ultimate source of the decision rather than merely the location at which a prior chain of physical causes terminates.

This raises the familiar question:

If a person's complete four-dimensional history contains deliberation followed by choice A, could that person have chosen B instead?

The answer depends on what is meant by could.

The Block Universe itself does not say that only one conceivable four-dimensional history exists. Consider two possible complete histories:

$$\text{BA:}\quad \text{Spre} \rightarrow \text{A}$$

$$\text{BB:}\quad \text{Spre} \rightarrow \text{B}$$

The two histories can be identical up to the decision event and differ only in the subsequent choice. Nothing about the Block Universe ontology itself prohibits such a pair of possible histories.

The Block Universe says that the actual universe is one complete four-dimensional history. It does not answer why the actual universe is BA rather than BB, nor does it by itself specify the physical laws or boundary conditions that determine which complete history is realized.

The Block Universe is not a theory of why this particular Block Universe exists rather than another possible Block Universe.

Consequently, the Block Universe should not be confused with determinism. Determinism is a claim about the dynamics of a physical theory. The Block Universe is an ontological description of spacetime.

$$\text{Block Universe} \neq \text{Deterministic Universe}$$

$$\text{Block Universe} \neq \text{Predetermined Universe}$$

A deterministic physical theory may imply that a particular prior state permits only one subsequent state. But that conclusion comes from the deterministic laws, not from the Block Universe concept itself.

Likewise, an indeterministic theory may permit multiple possible complete histories while the actual Block Universe contains only one of them.

5.5 Two meanings of “could have chosen differently”

The phrase “I could have chosen differently” therefore needs to be treated carefully. There are at least two distinct meanings.

5.5.1 Possible-history sense

There exists another physically possible complete history that is identical to the actual history up to the decision and in which the person chooses differently.

$$\text{BA:}\quad \text{Spre} \rightarrow \text{A}$$

$$\text{BB:}\quad \text{Spre} \rightarrow \text{B}$$

Under this interpretation, the Block Universe is entirely compatible with the statement that the person could have chosen differently.

5.5.2 Counterfactual-within-the-same-history sense

A stronger interpretation asks whether the person could have changed the actual outcome while leaving the actual four-dimensional history otherwise unchanged.

This is a substantially different proposition. The Block Universe does not require us to adopt it, and it is not necessary to establish that alternative complete histories are possible.

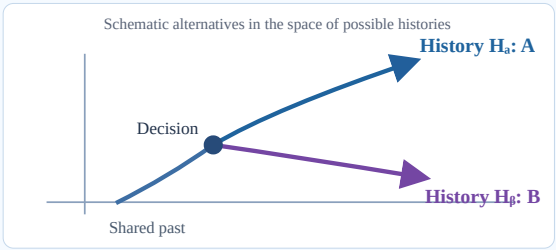
“Another possible history” is not the same thing as “another outcome within the same actual history.”

Once this distinction is made, the apparent contradiction between libertarian free will and the Block Universe becomes considerably less immediate.

Figure 5.1: Two Meanings of “Could Have Chosen Differently”

(a) Alternative Possible Histories

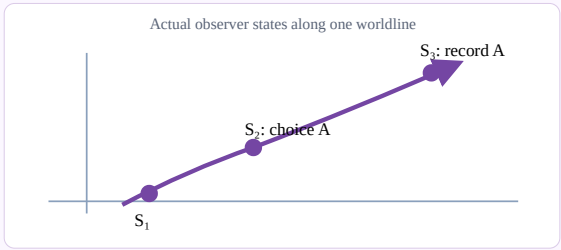
Different complete histories may share the same earlier physical conditions.



Alternative histories are possibilities, not simultaneous actual events.

(b) One Actual History

Only one choice is part of the actual complete spacetime history.



The actual history contains A, not A and B.

Key distinction “Another physically possible history” is not the same as “changing the outcome while holding the actual complete history fixed.”

5.6 Sourcehood and personhood

Another conception of free will focuses not on metaphysical indeterminacy but on sourcehood: whether the person is the genuine source of the action.

Every physical record in the universe has an antecedent history. The physical history of a human being can ultimately be traced backward through the history of life, the formation of the Earth, stellar and galactic evolution, and ultimately toward the earliest physical conditions of the universe.

Universe → galaxies → stars → planets → life → brains → deliberation → choice

But the existence of this larger causal history does not eliminate the person as a distinct physical and causal system.

A person's decision can simultaneously be described at different levels. It can be a physical event caused by the state of the person's brain, while also being an action caused by the person's beliefs, memories, preferences, intentions, and deliberation.

These descriptions need not compete with one another. The person is a physical structure within the larger causal structure of the universe.

In this sense, personhood and sourcehood can be understood as properties embodied along the person's world line. The fact that the world line has physical antecedents extending back through the universe does not erase the causal reality of the person at the scale at which the person's decisions and actions occur.

The existence of antecedent causes extending throughout the history of the universe does not eliminate the person as a distinct physical and causal system within that history.

At this point the question begins to move from physics into philosophy. Physics can describe the physical processes constituting the person and the causal relations among them. Whether that constitutes the kind of “ultimate authorship” required by a particular philosophical conception of free will is a further philosophical question.

5.7 Non-physical and dualist conceptions

A fundamentally different proposal is that conscious agency is not entirely physical. A soul, mind, or other non-physical agent could be proposed as an additional source of choice.

The Block Universe does not itself disprove such a proposal. The issue is different: without a theory describing the proposed non-physical entity and its interaction with the physical universe, the proposal does not provide a physical explanation of the choice.

For example, suppose we write:

$$\text{brain state} \rightarrow \text{non-physical agent chooses} \rightarrow \text{brain state changes}$$

We then need to ask what the state of the non-physical agent is, what rules govern its behavior, how it interacts with physical matter, and what observable consequences distinguish this account from a purely physical explanation.

Simply asserting that “the soul made the choice” relocates the unexplained mechanism rather than specifying it.

A non-physical theory could certainly be developed in principle, but it would require additional structure before it could function as an explanatory physical theory. Otherwise, sufficiently unconstrained metaphysical agency can be invoked to explain almost any outcome, making it difficult to distinguish the hypothesis empirically.

This limitation is not imposed by the Block Universe. It is a general requirement for constructing a testable physical explanation.

5.8 Free will and the independence of other conscious entities

Our original definition included another interesting condition: that an individual's deliberation should be independent of the internal deliberations of other conscious entities.

This should be distinguished from statistical correlation.

As established in Section 4, the existence of a correlation does not by itself imply a causal interaction. In particular:

correlation across spacetime ≠ interaction across spacetime

The same distinction applies to human decisions. If two people's decisions are correlated, that does not by itself establish that one person's deliberation has overridden or controlled the other's deliberation.

For free will, the more relevant question is whether another agent has bypassed or coerced the person's own deliberative process.

Thus statistical dependence should not automatically be identified with loss of agency. This is particularly important in the present framework, because the Block Universe permits correlations among events at different locations and times without requiring a corresponding physical signal propagating between them.

5.9 The Block Universe does not determine the origin of the actual history

We can now state an important limitation of the framework.

The Block Universe tells us that the actual universe is a complete four-dimensional history. It does not tell us why that particular history exists rather than another possible history.

Suppose two complete histories are physically possible:

$$BA = \{\text{all events, with choice A}\}$$

$$BB = \{\text{all events, with choice B}\}$$

The Block Universe framework does not contain an additional principle that says: “BA must exist rather than BB.”

That question belongs to whatever deeper theory specifies the physical laws, boundary conditions, probability structure, or other principles governing the universe.

This is analogous to the role of the Block Universe in our discussion of quantum mechanics. In Section 4, the Block Universe provided a framework in which complete four-dimensional histories could contain correlations without requiring instantaneous causal influence. It did not itself derive the particular quantum probability distribution

$$\rho(\lambda|x,y) = [1 - \lambda_1\lambda_2(\mathbf{x} \cdot \mathbf{y})] / 4.$$

Likewise, in the present discussion, the Block Universe provides the four-dimensional ontology but does not itself determine why a particular person's world line contains one choice rather than another.

5.10 A useful taxonomy of free will

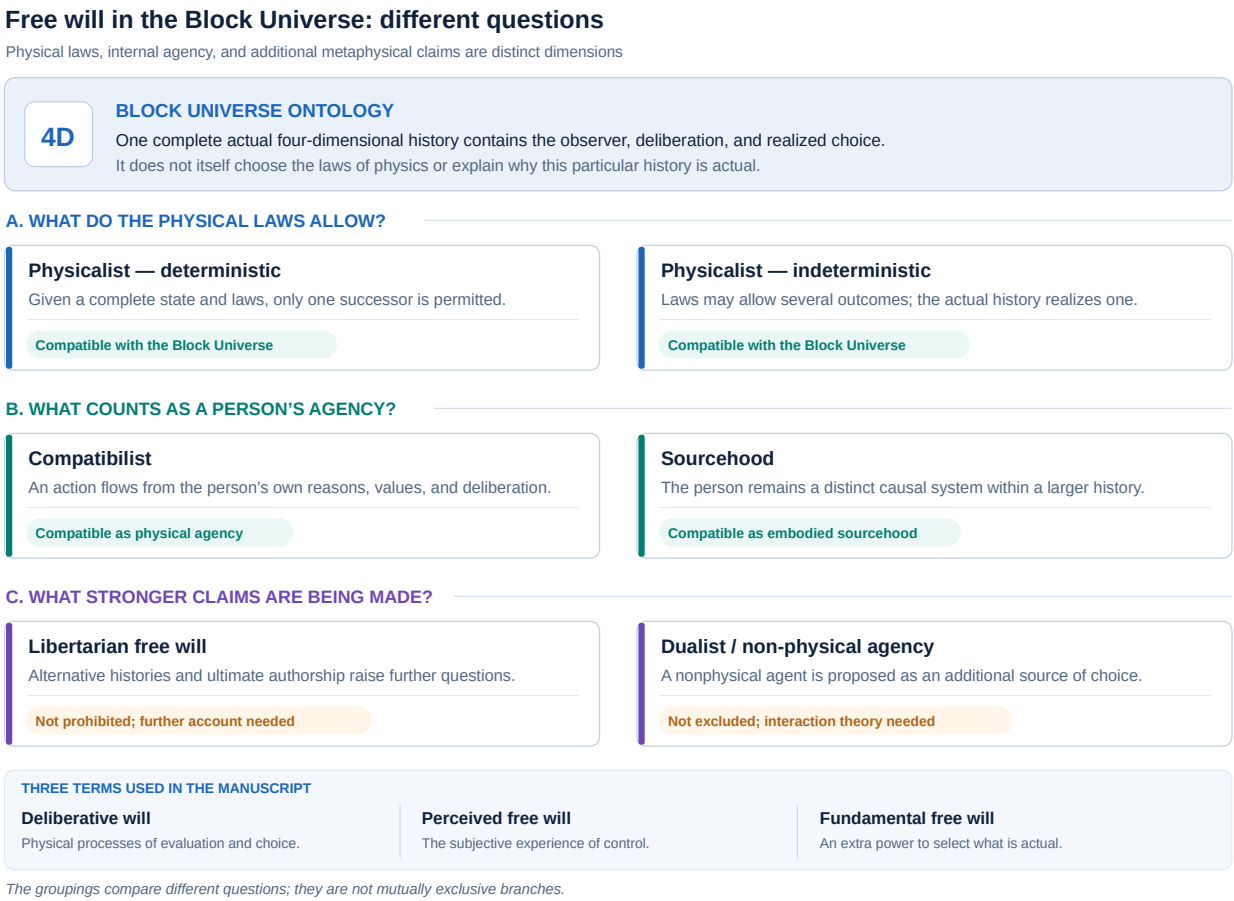


Figure 5.2. *A conceptual map of free-will interpretations.*

Deterministic and indeterministic laws are compatible with one actual four-dimensional history. Compatibilism and sourcehood describe agency within it. Libertarian ultimate authorship and nonphysical agency raise additional questions not settled by the Block Universe; the categories are not mutually exclusive.

Conception	What makes an action free?	Compatibility with the Block Universe
Physicalist-deterministic	Physical deliberation produces the action under deterministic laws.	Compatible. The Block Universe is indifferent to determinism.
Physicalist-indeterministic	Physical deliberation produces the action while the underlying laws contain genuine indeterminacy.	Compatible. The actual Block Universe contains one history permitted by the laws.
Compatibilist	The action flows from the person's own reasons, memories, preferences, intentions, and deliberation.	Compatible. The deliberative process is itself part of the person's world line.
Libertarian	The person is genuinely capable of alternative choices and is an appropriate ultimate source of the decision.	Not prohibited by the Block Universe. Alternative complete histories can be conceived without requiring the Block Universe to explain why the actual one is realized.
Sourcehood	The action originates through the person's own deliberative and motivational structure.	Compatible. A person remains a distinct causal system within the larger four-dimensional history.

Conception	What makes an action free?	Compatibility with the Block Universe
Dualist / non-physical	A non-physical component participates in or controls the decision.	Not ruled out by the Block Universe, but requires an additional theory describing the non-physical agent and its interaction with the physical universe.

5.11 What the Block Universe actually implies about free will

Under the definition used throughout this manuscript, the actual Block Universe is the complete physical history of reality: one particular realization consistent with the laws of physics. It contains the observer, the brain, the deliberation, any genuinely random events, and the realized decision. It is not a stage on which an additional agent stands outside the history and selects what will happen.

This makes the phrase independent control problematic unless its meaning is specified. Independent of what? If it means independent of the complete physical history or able to suspend physical laws, it is incompatible with the physicalist Block Universe being considered here. If it refers instead to a physically lawful process inside the history, that process is already part of the universe; it may be described as agency or regulation, but it does not introduce a separate freedom to choose which complete history is actual.

Neither determinism nor indeterminism changes this reasoning. Deterministic laws may permit one outcome given a complete prior state; indeterministic laws may permit several with specified probabilities. Randomness does not confer control over the random result upon the agent. In either case the actual Block Universe contains the outcome that occurred.

This does not make deliberation unreal or causally irrelevant. The agent's reasoning is part of the physical account of the realized action. Nor does it imply that the action was externally coerced: a person and the physical processes constituting that person are not two competing sources of causation.

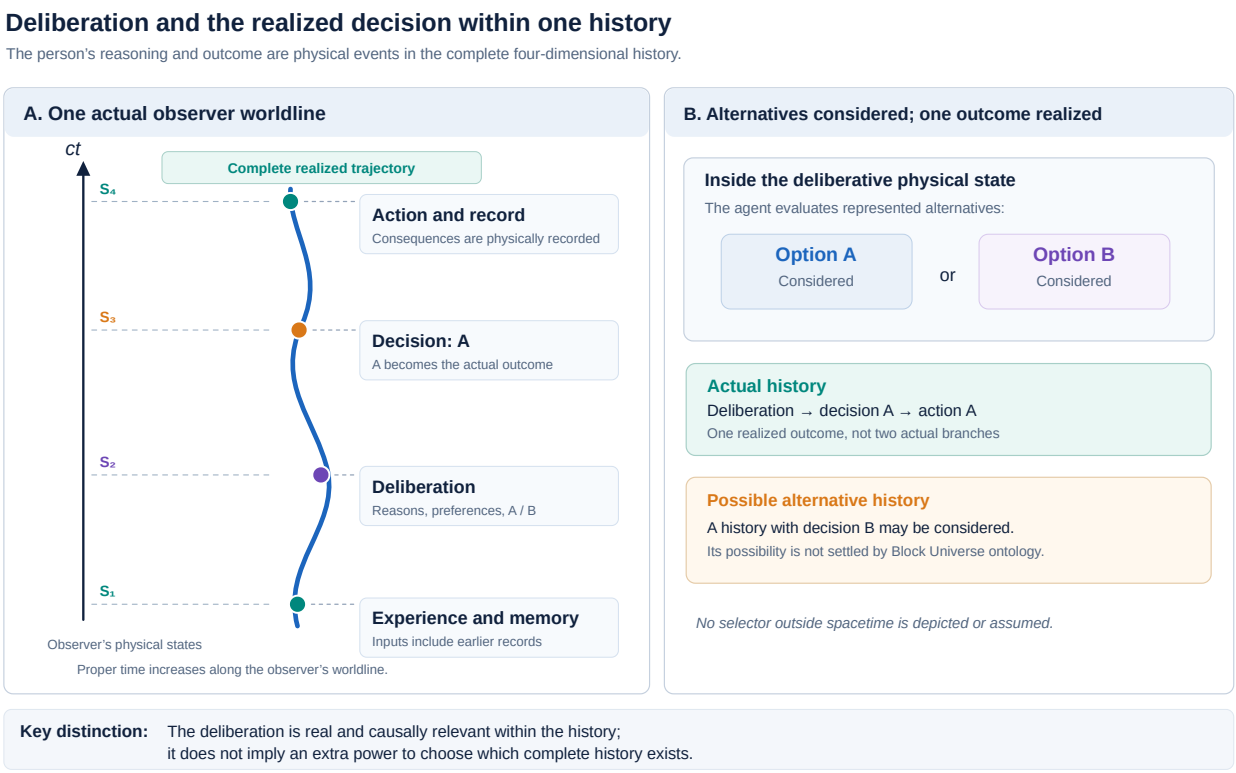


Figure 5.3. *Deliberation along an observer's worldline.*
The actual history contains memory, evaluation of alternatives, a realized decision, and subsequent action.
Represented alternatives are not additional realized branches; the Block Universe does not select why this history is actual.

The observer is part of the realized history, not an additional selector of that history. Deliberative will is physically real; the stronger freedom commonly attributed to it is not established by the experience of choosing.

Here, to say that the laws of physics govern the decision means that the complete history must satisfy physical constraints, not that the laws themselves are conscious agents that actively choose outcomes. The Block Universe does not supply an explanation of why this permissible history exists rather than another.

5.12 The central conclusion

The distinction between deliberative will and fundamental free will allows us to state the present interpretation more precisely. An observer may consciously consider A and B, revise intentions, and ultimately realize A. Every part of that process is contained in the physical history. The decision, in the sense used here, is the realized A rather than B; earlier tentative commitments are simply events within the deliberation leading to it.

The observer may experience the process as an independent act of choosing. From the observer's immediate perspective, that experience may be indistinguishable from what the observer calls genuine free will. But the experience alone does not establish an additional power to select or revise the complete physical history.

This is parallel to our account of temporal flow. Physical records underpin an observer's experience of time passing without requiring an objectively moving present. Similarly, the physical processes and records of deliberation and choice can underpin an experience of free will without requiring a separate selector of physical reality.

Our preferred interpretation is that deliberative will is a real physical capability, while perceived free will is a real subjective experience. Neither entails fundamental freedom from the physical history or its governing laws.

The conclusion does not depend on determinism. It also does not purport to derive the subjective experience of agency from a completed theory of consciousness. It is an ontological interpretation grounded in the physicalist Block Universe assumed here, not a theorem of relativity alone or a demonstration that every philosophical account of agency is incoherent.

Alternative philosophical definitions, including compatibilist and sourcehood accounts, can remain meaningful descriptions of internal physical agency. What this interpretation rejects is the inference from those descriptions to an extra power of selection outside the complete physical history.

6. The Block Universe and Time Travel

Backward travel, causal interactions with the past, logical self-consistency, and closed timelike curves

The central issue raised by backward time travel is not whether a traveler's worldline closes, but whether all events—including interactions with an earlier history—can belong to one logically consistent Block Universe.

6.1 Time Travel as a Question of Causal Structure

In special and general relativity, a massive observer follows a future-directed timelike worldline. Proper time increases along that worldline. Relativistic time dilation allows two observers to accumulate different amounts of proper time between meetings; this familiar effect is sometimes called travel into the future. The more challenging question is whether an observer can reach the chronological past of an event from which they departed.

Let A be the traveler's departure event and B the arrival event. The chronological past $I^-(A)$ consists of events from which a future-directed timelike path to A exists. In flat spacetime this is the interior of A's past light cone. Thus $B \in I^-(A)$ means that B can causally precede A by a timelike route. The causal past $J^-(A)$ also includes events connected to A by lightlike paths.

An arrival at a numerically smaller coordinate time is not enough to establish physical travel into the past. Coordinate labels can be changed. The invariant question is whether B lies in the causal or chronological past of A while the traveler reaches B by following a locally future-directed causal path from A. General relativity admits mathematical spacetimes with such nontrivial global causal relationships, although their physical realization remains uncertain.

Crucially, the traveler's own worldline need not close. The observer can depart at A, arrive at B, and continue onward without ever returning to the departure event. Backward time travel and a closed traveler worldline are therefore different concepts.

6.2 Open Worldlines and Encounters with the Earlier History

After arriving at B in the chronological past of A, the traveler can follow an open worldline. Several possibilities arise. The traveler might avoid any interaction with their younger self. They might send a message to their younger self. Or they might interact with other people and objects whose subsequent histories affect the circumstances of departure at A.

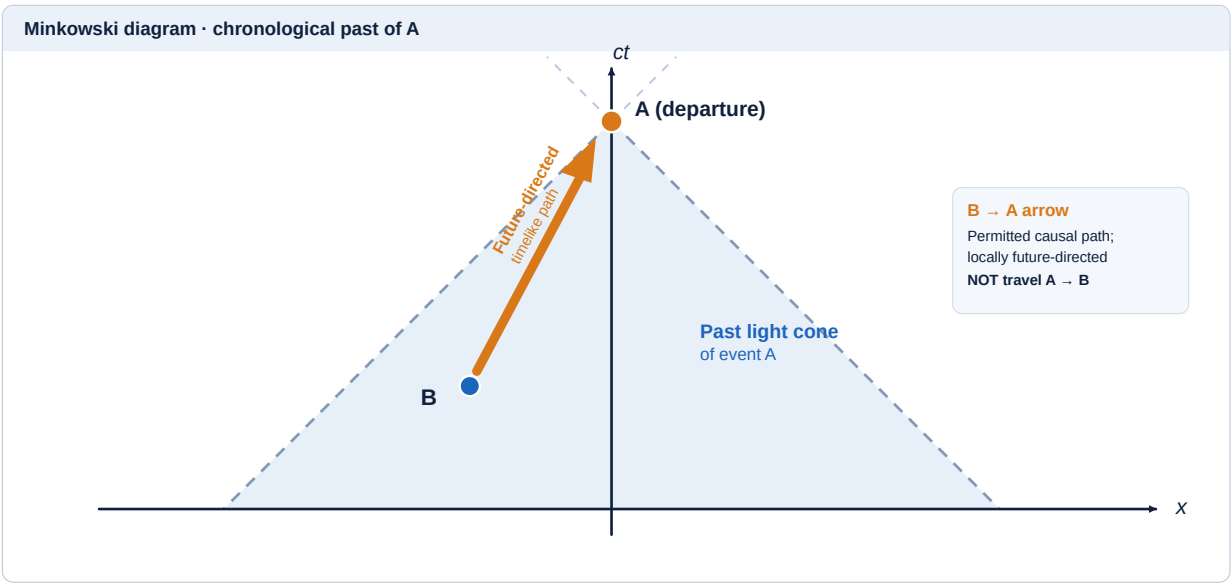
A meeting between older and younger versions of the same person does not require the older traveler to revisit the departure event A. The two versions can be present in the same region of spacetime at an event distinct from A, or communicate across distinct events. If they literally occupy the same event, the traveler's full worldline has a self-intersection there; it need not be a globally periodic worldline. A message exchanged across separate events can leave the material worldline non-self-intersecting.

The relevant condition is that the earlier person's memories and records, the later departure, and the older traveler's actions all agree as parts of one physical history. If the younger observer remembers receiving a warning, then the older observer who eventually sends it cannot belong to an otherwise identical history in which the warning was never received.

Even when there is no direct encounter, indirect causal effects may matter. Avoiding one's younger self removes an obvious opportunity for contradiction, but it does not independently prove the complete history consistent. Conversely, interaction with one's younger self is not itself paradoxical.

Causal geometry: B lies in A's chronological past

A future-directed timelike path from B to A does not depict travel backward in time



Key distinction: B lies in the chronological past of A; the orange arrow points toward increasing proper time.
The figure does not show how a traveler could depart at A and arrive at the earlier event B.

Figure 6.1. *The chronological past of departure event A.*

Event B lies within the past light cone of A, so a future-directed timelike path can connect B to A. The orange arrow shows this B → A causal path, not the traveler's proposed backward-time journey from A to B. The diagram is a schematic local spacetime representation; it does not establish the physical possibility of travel to the past.

6.3 The Grandfather Paradox Without a Closed Traveler Worldline

The grandfather paradox is an example of backward time travel with a potentially open traveler worldline. A traveler departs at A, arrives at B in an earlier era, and attempts to kill an ancestor before the ancestor has the descendants required for the traveler's own birth. The traveler need not return to A, and the observer's material trajectory need not form a closed loop.

The contradiction arises from the stipulated facts: the traveler exists and makes the journey, yet the intervention is specified to prevent the conditions necessary for that same traveler to exist. Both claims cannot be true of one complete history under the assumed genealogy and circumstances.

The paradox is often narrated as though a previously completed past were subsequently modified. In the Block Universe developed in Section 1, however, departure, arrival, intervention, and their consequences are all events in one four-dimensional history. There is no second temporal process in which an already existing past

is rewritten.

This shows why the grandfather paradox is more general than a paradox involving a traveler who returns to the same spacetime event. Its essential feature is a contradictory causal specification, not geometric closure of the observer's worldline.

6.4 Logical Self-Consistency as a Condition for Existence

The preceding examples motivate a principle broader than time travel: logical self-consistency is a necessary condition for physical existence. A complete history containing mutually contradictory facts cannot be an actual physical history.

Exists(H) ⇒ LogicallyConsistent(H).

The implication is one-way. Logical consistency does not establish that a hypothetical history exists, nor that it satisfies Einstein's field equations, quantum theory, conservation laws, or appropriate boundary conditions. A consistent imagined history may still be physically impossible.

Applied to the grandfather paradox, this principle does not posit a special force that jams the weapon or arranges a last-minute rescue. It states only that the contradictory set of events cannot constitute one actual history. A specific consistent history may contain a failed attempt, different ancestry, or some other outcome, but the principle by itself does not predict which outcome occurs.

The same requirement applies whether the traveler's worldline is open or closed, and whether causal contact with the younger self is direct, indirect, or absent. Time travel makes the constraint conspicuous; it does not create the requirement.

6.5 Permitted Causal Paths Versus Realized Physical Histories

General relativity describes a spacetime metric and the causal paths it permits. The actual Block Universe additionally contains the matter, radiation, observers, interactions, and physical records that are realized. A geometrically permitted trajectory that no object follows is not an extra material history or a second copy of an observer.

If the traveler goes from A to B along a future-directed timelike route while B lies in $I^-(A)$, there must also be a future-directed timelike route from B to A. Concatenating these routes establishes that the geometry admits a closed timelike path. This conclusion does not require the traveler to take the return route.

Thus an open actual traveler worldline can coexist with closed timelike curves admitted by the spacetime geometry. The existence of a geometric curve is not the same claim as the existence of a physical observer traversing it. The distinction matters when interpreting general relativity, but the ontological analysis of paradoxes concerns the actual events and their consistency.

Where an actual traveler affects their own earlier history, the relevant causal relationships can constrain that history without requiring the traveler's own worldline to close.

6.6 Closed Timelike Curves as a Special Geometrical Case

A closed timelike curve (CTC) is a future-directed timelike curve that returns to the same spacetime event. Writing the curve as $\gamma(\tau)$, with τ proper time, its geometric closure can be represented by

$$\gamma(\tau + T) = \gamma(\tau),\ T > 0.$$

Proper time increases locally throughout the circuit. The curve is globally closed even though the observer never reverses their own local temporal direction or exceeds the speed of light.

Gödel's rotating-universe solution provides a classic mathematical example. Certain idealized Kerr extensions and hypothetical time-shifted wormhole constructions also motivate discussions of chronology violation. These geometries are not evidence that realistic time machines exist.

The conceptual example A → B → A can be read in two ways. If the traveler actually takes both future-directed segments, the full path is closed. If the traveler takes only A → B and proceeds elsewhere, their path remains open, although a geometrically permitted B → A route may still exist. In neither case should a merely permitted return route be mistaken for an additional physical traveler.

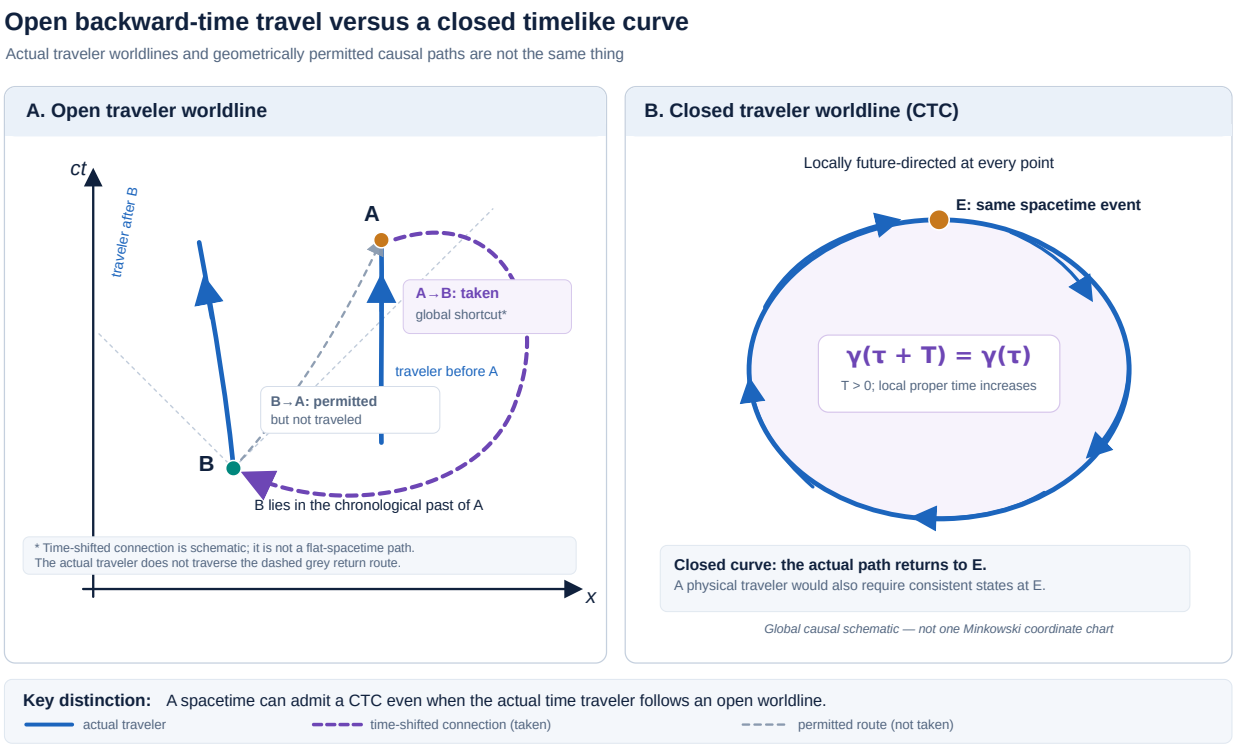


Figure 6.2. *Open backward-time travel versus a closed timelike curve.*
(A) A future-directed journey from departure A to earlier B through nontrivial global geometry is part of an open traveler worldline. A separate geometrically permitted B→A return route is not taken, even though together the paths establish the existence of a closed timelike path. (B) A closed traveler worldline returns to the same event while proper time increases locally around the loop. Such a realized trajectory additionally requires a self-consistent physical state; these are schematic causal structures, not evidence of a working time machine.

6.7 A Closed Worldline and the Traveler's Physical State

Geometric closure identifies the same spacetime event after a circuit, but a physical traveler also has internal degrees of freedom: a clock reading, memories, biological state, and interactions with the environment. These cannot be assigned arbitrarily to different traversals of the same event.

For a single-valued physical state S transported around a genuinely periodic worldline, consistency requires

$S(\tau + T) = S(\tau)$.

Equivalently, if F_T represents the state evolution around a circuit, a consistent periodic state is a fixed point: $F_T(S) = S$. A given physical evolution law need not possess such a state. Geometry alone does not guarantee a realizable material traveler.

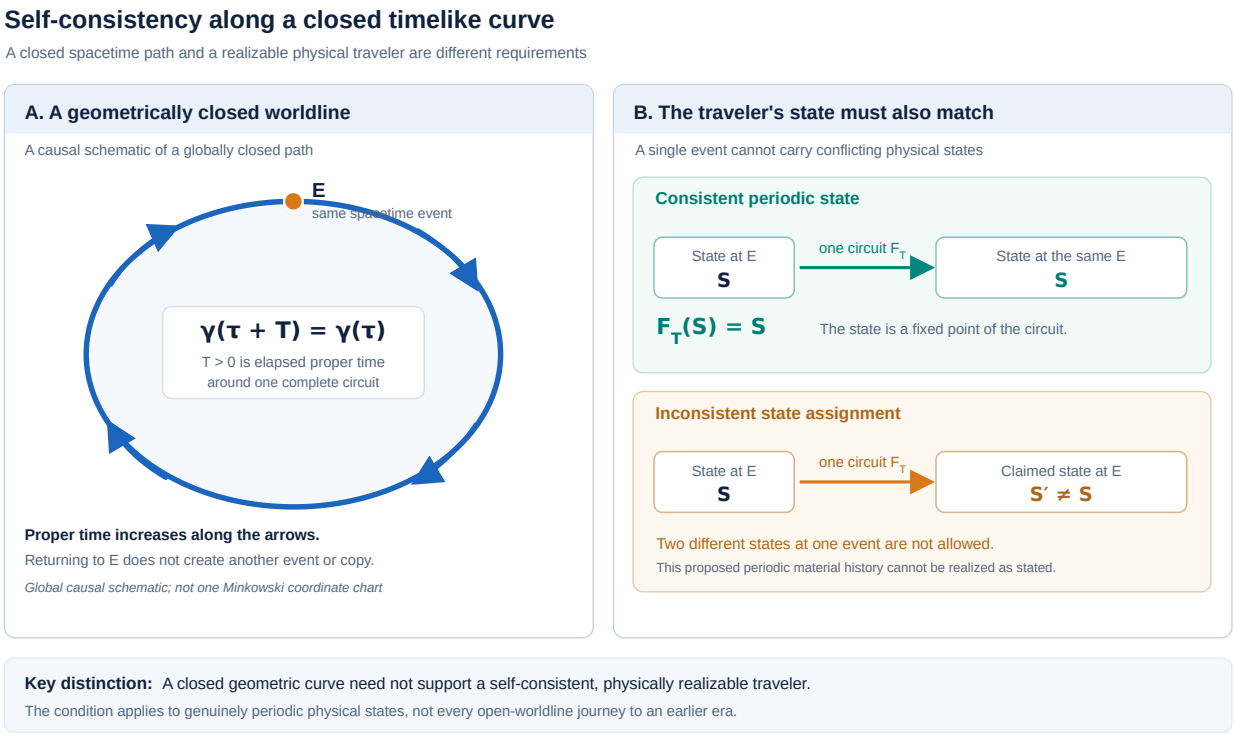


Figure 6.3. *State consistency along a closed timelike curve.*
(A) A future-directed timelike worldline returns to the same spacetime event E after proper time $T > 0$. (B) A realized traveler on a genuinely periodic path must have a single-valued internal state S, requiring $F_T(S) = S$. A mismatched return state would be inconsistent; geometric closure alone does not establish that a physical traveler can complete the circuit.

This explains why an ordinary-looking B → A journey can seem to create two copies at A. In the idealized smooth periodic case, the returning and departing portions are the same locally identified history, not two independently specified bodies that must occupy the same event with conflicting states. A self-intersecting but nonperiodic trajectory, or a model with multiple streams meeting, requires a more detailed matter description; geometric coincidence alone does not identify all internal states.

The state-matching condition is therefore a strong constraint on a realized closed physical trajectory. It is not a requirement that every open-worldline traveler returning to an earlier era become a periodic physical system.

6.8 Proper Time, Memory, and the Experienced Direction of Time

Section 2 explained the experienced direction of time in terms of asymmetric physical records. An observer remembers events represented in earlier physical states but does not ordinarily possess corresponding records of later states. This information asymmetry is linked to thermodynamic conditions, rather than to an external movement of a universal present.

An observer on an open backward-time journey continues to experience increasing proper time. The same local statement applies along a CTC. In either case, a journey to the chronological past of departure need not feel like time running backward.

Whether an observer could detect the global closure of their own worldline is a separate question. Some spacetime configurations may offer observable evidence of unusual causal structure; others may not. The mere subjective experience of a succession of remembered states does not establish that a worldline is globally open.

One must not assume that memory information strictly increases at every point along every worldline. Forgetting, erasure, equilibration, and exchange with the environment are physical possibilities. Nonetheless, the requirement of a globally consistent state around a realized periodic curve can be highly restrictive for an ordinary memory-forming organism.

6.9 Thermodynamics and Recurrent Physical States

Ordinary biological activity and memory formation typically involve dissipative processes. If a physically isolated system is required to return to precisely the same state after one circuit, then a strictly increasing single-valued entropy around that circuit is incompatible with closure.

$S_{entropy}(\tau + T) = S_{entropy}(\tau)$,

whereas

$dS_{entropy}/d\tau > 0$

everywhere would imply a contradiction.

This is a conditional incompatibility, not a general theorem excluding every CTC. Thermodynamic entropy increase is statistical; local decreases and transfers of entropy between subsystems and environments must be treated within the full physical model. Nor is there necessarily a globally defined time coordinate or universal entropy function that increases on every causal trajectory in a chronology-violating spacetime.

A closed parameterized curve may be traversed repeatedly in its mathematical description: $\gamma(\tau + nT) = \gamma(\tau)$. These repeated parameter values do not, by themselves, create infinitely many new spacetime events, independent copies of the traveler, or repeated deposits of heat. Distinct matter streams or physical interactions, if present in a proposed model, must still be accounted for.

The appropriate conclusion is modest: thermodynamics places additional constraints on physically realizable closed trajectories, particularly those involving ordinary observers, but it does not by itself convert every proposed time-travel history into a logical contradiction.

6.10 Global Consistency Without a Dynamical Intervention

Section 3 described physical laws as global relationships satisfied by a complete four-dimensional history, rather than necessarily as step-by-step instructions executed by an external clock. This viewpoint makes the time-travel puzzle less mysterious.

If the actual Block Universe contains an older traveler sending a message to a younger self, the reception, the later memories, the departure, and the transmission are all parts of one history. Nothing is added to a previously finished past. A contradictory description is not a real history awaiting repair.

This resembles the interpretation proposed in Section 4 for globally correlated quantum measurement settings and outcomes. In that discussion, global correlations need not be generated by superluminal signals or by a mechanism anticipating future choices. Here, consistent relationships between earlier and later events similarly need not be produced by a universe that detects paradoxes and intervenes to prevent them.

The analogy is conceptual, not a derivation. The Block Universe does not calculate which consistent time-travel history occurs, establish the dynamics of a time machine, or derive the statistical predictions of quantum mechanics. It offers an ontological interpretation of histories that are otherwise physically admissible.

6.11 Physical Possibility and the Central Conclusion

General relativity admits mathematical spacetimes with nontrivial causal structure, including closed timelike curves. Whether realistic matter and quantum effects permit the formation or traversal of such regions remains unresolved. The logical self-consistency principle neither proves time travel possible nor substitutes for the dynamical laws that any proposed geometry and matter configuration must obey.

The grandfather paradox illustrates the central point without requiring the traveler’s own worldline to close. Backward time travel may create causal relationships with an observer’s earlier history, and those relationships must form one consistent set of physical facts. The traveler may interact with a younger self, influence earlier events indirectly, or avoid such interactions; none of these alternatives escapes the general requirement of logical consistency.

Closed timelike curves are a special geometric manifestation of chronology violation. A spacetime can admit them without any observer actually traversing one, while an observer who does follow a genuinely closed physical worldline faces additional state-matching and thermodynamic constraints.

> The Block Universe does not portray time travel as rewriting the past. It portrays any physically realized journey, its causes, its interactions, and its consequences as parts of a single four-dimensional history. Logical consistency is necessary for that history to exist; physical laws govern which consistent histories, if any, are physically realizable.

Appendix A. CHSH and the Explicit Measurement-Dependent Local Hidden-Variable Model

This appendix contains the technical development underlying the discussion in Section 4. It is intentionally separated from the main narrative so that the conceptual argument can proceed without requiring the reader to follow the full CHSH algebra. The appendix derives the CHSH bound for local hidden-variable models, identifies the precise role of measurement independence, and presents the explicit measurement-dependent local hidden-variable model used in the main text.

The appendix supplies the mathematical support for the conclusions summarized in Section 4. The explicit model is a demonstration of mathematical possibility, not a complete physical account of how the relevant correlations arise. The Block Universe interpretation is distinguished below from the additional measurement-dependent hidden-variable proposal.

A.1 CHSH and Its Assumptions

The CHSH argument has two distinct ingredients. Bell-local response factorization, for a sufficiently specified hidden variable λ, takes the form

$$P(A,B|x,y,\lambda) = P(A|x,\lambda)P(B|y,\lambda).$$

Measurement independence is the separate statistical assumption

$$P(\lambda|x,y) = P(\lambda).$$

The familiar local hidden-variable CHSH bound follows when the relevant assumptions, including both factorization and measurement independence, hold. Neither assumption should be defined as the other. The deterministic derivation below identifies exactly where the common hidden-variable distribution enters.

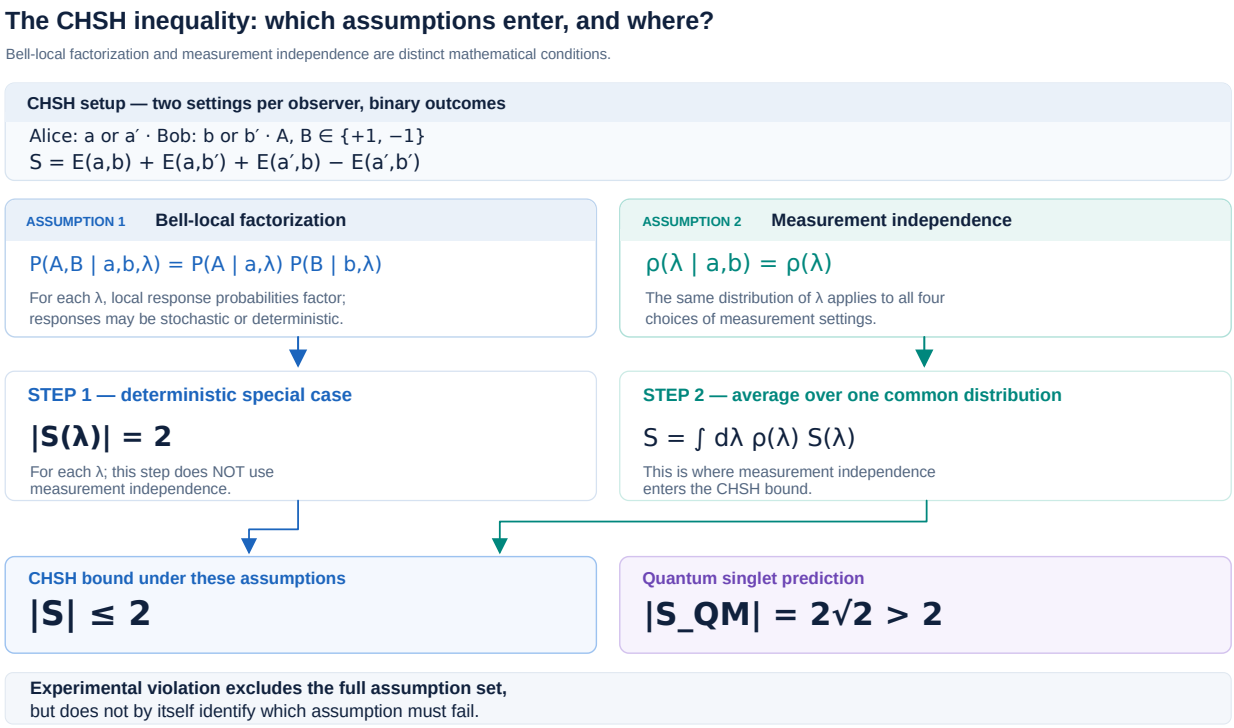


Figure A.1. *Logical structure of the CHSH inequality.*

For binary ±1 outcomes, the local deterministic algebra gives $|S(\lambda)| = 2$ for each hidden-variable state without using measurement independence. That assumption enters when all four correlations are averaged using the same distribution $\rho(\lambda)$, yielding $|S| \leq 2$. Quantum singlet correlations can reach $|S| = 2\sqrt{2}$, excluding models satisfying the full set of CHSH assumptions, but not isolating a unique assumption to reject.

A.1.1 The CHSH experiment

The CHSH formulation makes the logical structure of Bell's theorem especially transparent. Instead of considering a single measurement setting at each location, Alice has a choice between two settings, a and a', while Bob has a choice between two settings, b and b'. Each measurement produces an outcome of either +1 or −1.

For any pair of settings, define the correlation

$E(a,b) = \langle AB \rangle$.

The CHSH combination is then

$S = E(a,b) + E(a,b') + E(a',b) - E(a',b')$.

The remarkable feature of CHSH is that a local hidden-variable model satisfying measurement independence places a strict upper bound on this quantity.

A.1.2 The local deterministic bound

Begin with the deterministic special case:

$A = A(a,\lambda), \quad B = B(b,\lambda),$

where every outcome is either +1 or −1.

For a particular value of λ, define

$$S(\lambda) = A(a,\lambda)B(b,\lambda) + A(a,\lambda)B(b',\lambda) + A(a',\lambda)B(b,\lambda) - A(a',\lambda)B(b',\lambda).$$

Factor this expression:

$$S(\lambda) = A(a,\lambda)[B(b,\lambda)+B(b',\lambda)] + A(a',\lambda)[B(b,\lambda)-B(b',\lambda)].$$

Because *B(b,λ)* and *B(b′,λ)* can each only be +1 or −1, one of the two bracketed terms must be zero and the other must be either +2 or −2. Therefore

$$|S(\lambda)| = 2.$$

This is an important point: the local algebra itself does not yet use measurement independence. The bound arises for each individual hidden-variable state.

A.1.3 Where measurement independence enters

Under measurement independence, the same probability distribution *ρ(λ)* is used for all four setting combinations:

$$E(a,b) = \int d\lambda\, \rho(\lambda)\, A(a,\lambda)B(b,\lambda),$$

$$E(a,b') = \int d\lambda\, \rho(\lambda)\, A(a,\lambda)B(b',\lambda),$$

$$E(a',b) = \int d\lambda\, \rho(\lambda)\, A(a',\lambda)B(b,\lambda),$$

$$E(a',b') = \int d\lambda\, \rho(\lambda)\, A(a',\lambda)B(b',\lambda).$$

Because the same distribution appears in all four expressions, they can be combined:

$$S = \int d\lambda\, \rho(\lambda)\, S(\lambda).$$

Since $|S(\lambda)| = 2$ for every λ, it follows that

$$|S| \le 2.$$

This is the CHSH inequality.

The exact point at which measurement independence enters is therefore extremely specific. It is **not** the algebraic identity that gives $|S(\lambda)|=2$. It is the assumption that the four experimentally different setting combinations can all be represented as averages over the **same underlying distribution** *ρ(λ)*.

If instead the distribution depends on the settings, we have

$$\rho(\lambda|a,b),\quad \rho(\lambda|a,b'),\quad \rho(\lambda|a',b),\quad \rho(\lambda|a',b'),$$

and the four terms cannot in general be combined into a single integral of *S(λ)*. The step producing $|S|\le 2$ is then unavailable.

Measurement independence → one common ρ(λ) → CHSH bound.

This is the mathematical opening on which the remainder of the proposal rests.

A.1.4 Deterministic notation and stochastic physics

At this stage it is important not to overinterpret the notation *A(a,λ)* and *B(b,λ)*. The deterministic treatment is mathematically convenient because it makes the CHSH logic transparent. The broader proposal does not require that nature be deterministic at the hidden-variable level.

A more general local stochastic model would instead specify conditional probabilities such as

$P(A|a,\lambda), \quad P(B|b,\lambda).$

Locality can then be represented by the factorization

$P(A,B|a,b,\lambda) = P(A|a,\lambda)P(B|b,\lambda).$

Measurement dependence concerns the relationship between λ and the settings, not whether the response functions themselves are deterministic. The proposal therefore remains compatible with genuinely stochastic underlying physics.

A.2 Quantum Predictions and Experimental Violations

A.2.1 The singlet prediction and CHSH violation

For a pair of spin-½ particles prepared in the singlet state, quantum mechanics predicts the correlation

$E_{QM}(a,b) = -\cos(\theta_{ab}),$

where θ_{ab} is the angle between the measurement directions.

Choose, for example,

$a = 0°, \quad a' = 90°, \quad b = 45°, \quad b' = -45°.$

The corresponding quantum correlations give

$|S_{QM}| = 2\sqrt{2}.$

Since

$2\sqrt{2} > 2,$

the quantum prediction violates the CHSH bound.

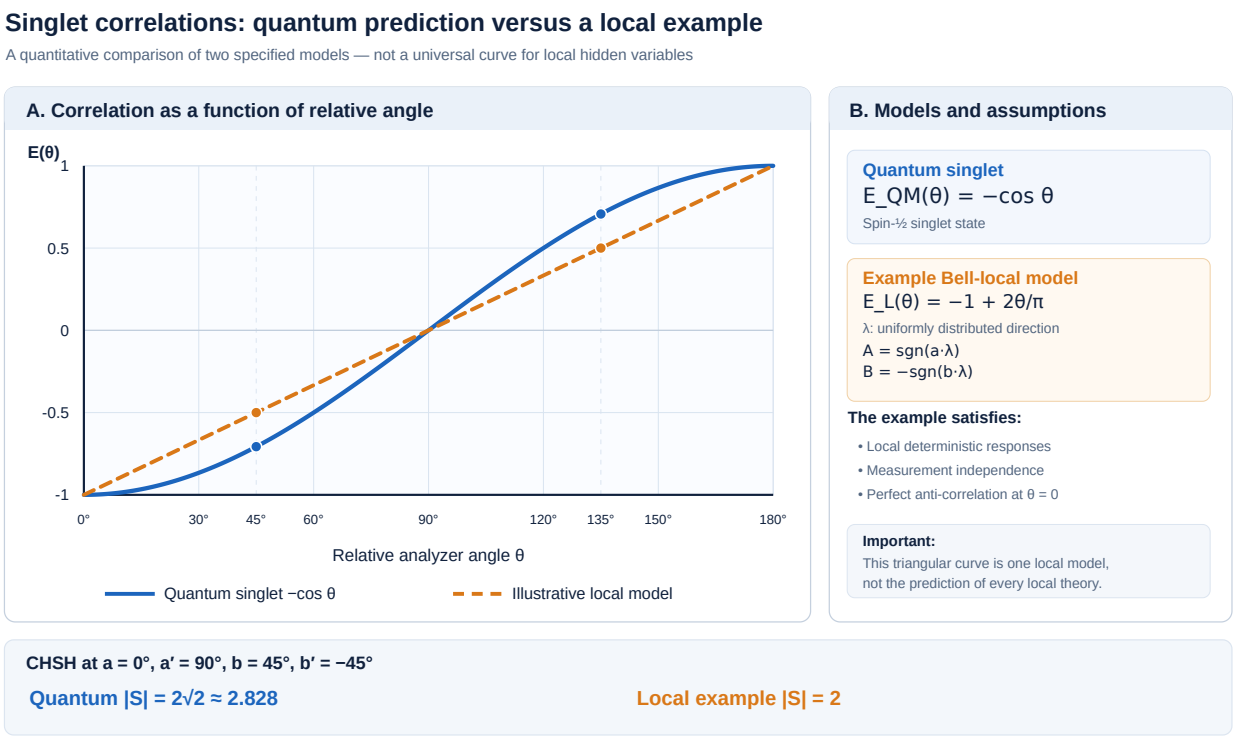


Figure A.2. *Quantum singlet correlations and one Bell-local, measurement-independent example.*

(A) The singlet predicts $E(\theta) = -\cos\theta$; a specified local model with a uniformly random hidden direction predicts $E(\theta) = -1 + 2\theta/\pi$ for $0 \le \theta \le \pi$. This comparison does not define a unique "classical" curve. (B) For the settings used in Appendix A.2, the quantum CHSH magnitude is $2\sqrt{2}$, while the illustrative local model reaches 2. The figure compares predictions, not mechanisms.

The experimental significance is straightforward: measurements reproduce correlations incompatible with the assumptions used to derive the local, measurement-independent CHSH inequality.

Quantum predictions and Bell-test observations violate the CHSH bound obtained under the complete set of relevant assumptions. The violation therefore excludes that class of models; it does not, by itself, isolate measurement independence as the assumption nature must reject. Nor does the violation demonstrate that a usable faster-than-light signal has been sent.

A.3 An Explicit Measurement-Dependent Construction

A.3.1 The constructive question

The CHSH analysis has established the logical opening, but a logical opening is not yet a physical model. To make the proposal more substantial, we need to show that a measurement-dependent local hidden-variable construction can actually reproduce the quantum correlations.

Such constructions are known in the foundations of physics literature. A particularly simple example, associated with Brans and discussed explicitly by Michael Hall, uses a hidden variable with two binary components. The model is useful here not because it constitutes the complete physical theory being proposed, but because it demonstrates that relaxing measurement independence is mathematically sufficient to construct local response functions reproducing the singlet statistics.

A.3.2 The hidden variable and local responses

Let the hidden variable be

$\lambda = (\lambda_1, \lambda_2)$,

where

$\lambda_1, \lambda_2 \in \{+1,-1\}$.

There are therefore four possible values of λ.

Define the local measurement responses simply as

$A(x,\lambda) = \lambda_1, \quad B(y,\lambda) = \lambda_2$.

These response functions are manifestly local in the relevant sense: Alice's result depends on her local hidden-variable component and not on Bob's measurement direction, while Bob's result depends on his local hidden-variable component and not on Alice's measurement direction.

A.3.3 The conditional distribution

The essential ingredient is the distribution assigned to the hidden variable. Instead of assuming a setting-independent distribution, define

$\rho(\lambda|x,y) = [1 - \lambda_1\lambda_2(x\cdot y)] / 4$.

Here *x* and *y* are the two measurement directions, represented as unit vectors.

This equation contains the entire departure from measurement independence. The local response functions remain simple. All of the angular dependence required to reproduce the quantum correlation is placed in the conditional distribution of the hidden variable.

In this example the response functions contain no dependence on the other party's setting. The angular dependence appears instead in the conditional hidden-variable distribution. That is the specific trade made by the construction.

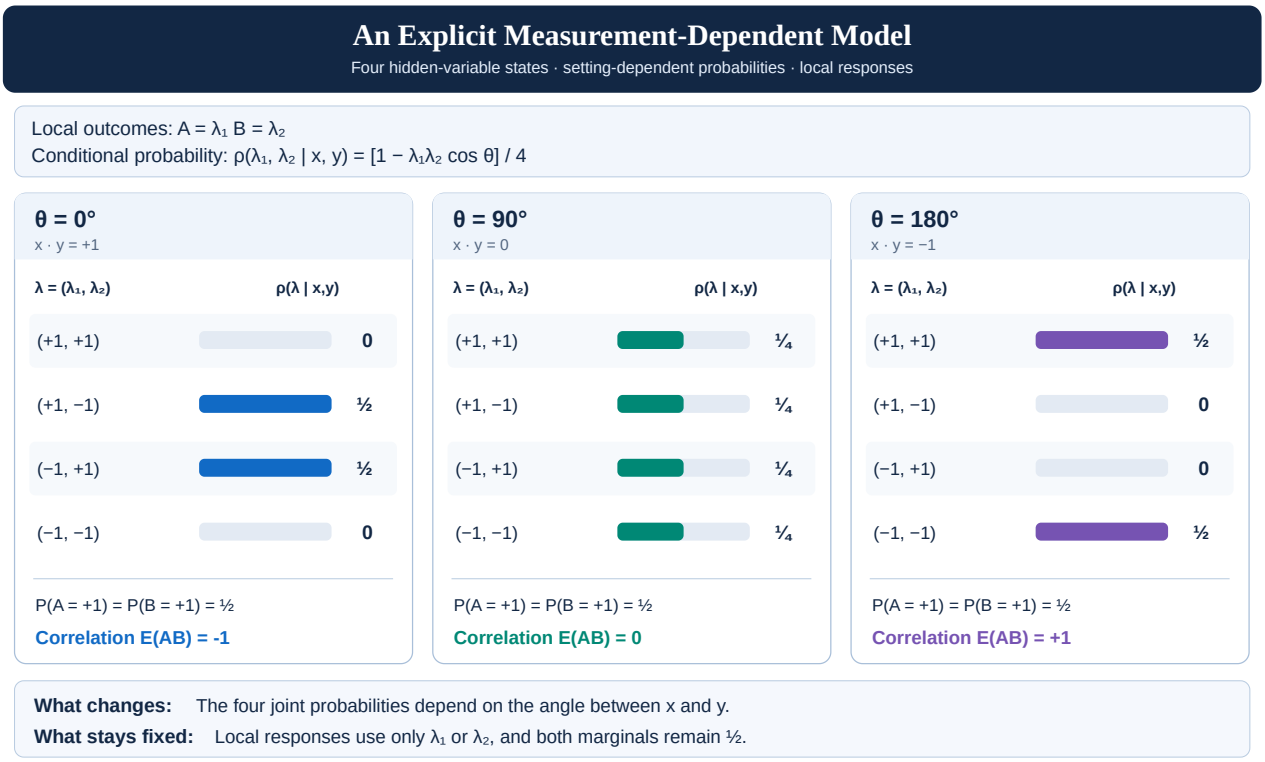


Figure A.3. *The conditional probabilities of a measurement-dependent local hidden-variable model.*
This construction reproduces singlet correlations for these angles, but does not explain the physical origin of measurement dependence. A single trial has one recorded outcome; columns describe ensembles.

A.4 Verification of the Model

A.4.1 Normalization and non-negativity

A probability distribution must be non-negative and must sum to one.

Because $\lambda_1\lambda_2=\pm 1$ and $-1 \le x\cdot y \le 1$, the numerator satisfies

$$0 \le 1 - \lambda_1\lambda_2(x\cdot y) \le 2.$$

Thus

$$\rho(\lambda|x,y) \ge 0.$$

Summing over the four possible values of λ gives

$$\Sigma\lambda\ \rho(\lambda|x,y) = [4 - (x\cdot y)\Sigma_\lambda \lambda_1\lambda_2]/4 = 1,$$

because the four values contain equal numbers with $\lambda_1\lambda_2=+1$ and -1.

The construction is therefore a valid conditional probability distribution.

A.4.2 Local marginal probabilities

A crucial property of the model is that Alice's marginal statistics remain independent of Bob's setting, and vice versa.

Since Alice's outcome is $A=\lambda 1$, sum over $\lambda 2$:

$$P(A=+1|x,y) = 1/2, \quad P(A=-1|x,y) = 1/2.$$

Likewise,

$$P(B=+1|x,y) = 1/2, \quad P(B=-1|x,y) = 1/2.$$

Thus, neither observer can infer the distant measurement setting merely by examining the local outcome statistics.

This is important conceptually. The model is measurement-dependent, but the dependence does not provide an operational faster-than-light signaling channel.

A.4.3 The singlet correlation

Because the outcomes are

$$A = \lambda_1, \quad B = \lambda_2,$$

their product is

$$AB = \lambda_1\lambda_2.$$

The predicted correlation is therefore

$$E(AB|x,y) = \Sigma\lambda\ \rho(\lambda|x,y)\ \lambda_1\lambda_2.$$

Substituting the conditional distribution gives

$$E(AB|x,y) = \Sigma\lambda\ [1 - \lambda_1\lambda_2(x\cdot y)]\ \lambda_1\lambda_2/4.$$

The first term sums to zero because the four hidden-variable states contain equal numbers of positive and negative products. The second term gives

$$E(AB|x,y) = -x\cdot y.$$

For unit measurement vectors this is exactly

$$E(AB|x,y) = -\cos(\theta_{xy}),$$

which is the singlet correlation predicted by quantum mechanics.

A.4.4 The full singlet joint probabilities

The construction is stronger than reproducing only the average correlation. Since $A=\lambda_1$ and $B=\lambda_2$, the joint probability of outcomes $a,b\in\{\pm 1\}$ is

$$P(A=a,B=b|x,y) = [1 - ab(x\cdot y)]/4.$$

This is precisely the singlet-state joint probability distribution for spin measurements along directions x and y.

Thus, the simple model does not merely produce the correct value of *E(AB)*. It reproduces the complete two-outcome probability distribution for the singlet measurement.

No-signaling in the constructed statistics. The local marginal probabilities above are uniformly 1/2 and do not depend on the distant setting. Thus, for the modeled singlet experiment, the observable outcome statistics give neither party a channel for inferring the other's setting. This verifies operational no-signaling for this distribution; it does not independently establish a causal mechanism that realizes the setting dependence.

A.5 What the Construction Establishes

The model demonstrates the following exact mathematical statement: **local response functions together with an appropriately setting-dependent hidden-variable distribution reproduce the singlet joint probabilities.** In particular, Bell-violating correlations need not be represented by response functions in which Alice's outcome explicitly depends on Bob's chosen setting or vice versa.

It is important to use the Bell terminology precisely. The conditional outcome probabilities in this example factorize into local responses, while the hidden-variable distribution violates measurement independence. These are distinct conditions. The model is therefore properly called a **measurement-dependent local hidden-variable model**, or a model with local response functions and setting-dependent hidden-variable statistics; it is not a model satisfying all assumptions needed for the usual CHSH bound.

There is a further limit to the construction's explanatory force. With unrestricted measurement dependence, a model of the same mathematical form could encode *any* target pair-outcome distribution Q(a,b|x,y): set $\lambda = (\lambda_1,\lambda_2)$, take $A=\lambda_1$ and $B=\lambda_2$, and choose $\rho(\lambda|x,y)=Q(\lambda_1,\lambda_2|x,y)$. The target distribution is then reproduced by construction. The singlet example is an exact and useful existence demonstration, but this flexibility does not explain *why* nature should realize that particular distribution.

Nor does a local-response factorization, by itself, constitute a complete physical account of local causation. The hidden variables, their relation to the measuring devices, and the origins of the setting-dependent statistics would have to be specified by a physical theory. The construction does not show that measurement independence is violated in nature.

The experiments remain unchanged: observed correlations violate the standard Bell inequalities, and quantum mechanics predicts those correlations. The appendix does not dispute either point. It shows mathematically what becomes possible when one assumption is relaxed, without providing a new experimentally established account of the correlations.

A.6 The Block Universe Interpretation

A.6.1 Correlations within a complete history

In the Block Universe interpretation developed in Sections 1–4, the source preparation, hidden variable, measurement settings, outcomes, observer states, and measurement records belong to one complete four-dimensional physical history. Schematically,

H = {spacetime geometry, preparation, λ, x, y, A, B, records, observer states, ...}.

A statistical dependence between λ and the settings need not, merely by being a dependence, establish that a later setting sends a physical signal into the past or that one distant result dynamically changes the other. **Statistical dependence is not itself a causal mechanism.** Which causal explanation, if any, applies to a particular dependence is a separate physical question.

The Block Universe does not imply that measurement independence fails. It provides an ontological context in which a proposed measurement dependence can be considered as a relation among events without assuming a new instantaneous or backward-propagating physical influence.

A.6.2 Joint probabilities and possible histories

The statistical relationship can be described by a joint distribution ρ(λ,x,y). The measurement-independence assumption is the factorization

$$\rho(\lambda,x,y) = \rho(\lambda)\rho(x,y).$$

The present proposal permits a joint distribution that does not factorize in this way. Whenever the relevant conditionals are defined, the statistical dependence may equivalently be written $P(\lambda|x,y) \neq P(\lambda)$ or $P(x,y|\lambda) \neq P(x,y)$. Neither conditional form alone specifies a direction of causation.

These distributions describe possible experimental outcomes and their statistical relationships. The actual Block Universe contains the particular events and outcomes that occur. A probability assignment over possible histories must not be confused with a claim that all those alternative histories are physically realized or that the Block Universe itself selects one of them.

A.6.3 Local dynamics and global constraints

Local physical laws describe relationships among physical quantities in spacetime neighborhoods. A proposed global constraint or boundary condition could, in principle, place additional restrictions on which complete histories satisfy those laws. This is a conceptual possibility, not a derivation of the singlet distribution from the Block Universe.

Logical consistency is necessary, but not sufficient, for physical realizability. Global consistency alone neither specifies probabilities nor determines which admissible history is actual. A viable physical model would have to supply the additional laws, boundary conditions, or statistical principles that produce the required correlations.

The result is an interpretive distinction: the complete four-dimensional history may be described as satisfying physical relationships throughout spacetime without treating those relationships as instructions executed by an external time. This perspective does not abolish ordinary local dynamics, and it does not establish that a particular nonlocal-looking statistical correlation has a local causal explanation.

A.7 Randomness, Measurement Choices, and Causality

A.7.1 Ontic randomness and observer records

Statistical dependence should not be confused with determinism. An underlying variable can be genuinely random and also statistically correlated with other physical variables. The example uses deterministic response functions for mathematical clarity, but does not require that the entire underlying physical history be generated by deterministic laws.

Probabilities may characterize alternative possible experimental outcomes under specified preparations and settings, while an individual trial has one definite recorded result. Similarly, one may consider probability assignments over possible complete histories without asserting that all are actual. The Block Universe ontology by itself does not determine the origin or probability of one particular actual history.

An observer encounters physical states and records sequentially along a worldline. Earlier observer states do not ordinarily contain usable records of later measurement outcomes. The four-dimensional description does not imply foreknowledge, a reversal of experienced time, or a physical change to an already completed past.

A.7.2 Measurement settings and free will

Measurement independence is a statistical condition on the relationship between λ and the settings chosen by Alice and Bob. It is not itself a philosophical definition of free will. Conversely, statistical dependence between physical settings and hidden variables does not, on its own, show that experimenters lack agency or have been coerced. Section 5 examines those separate questions directly.

A.7.3 Explanations, objections, and terminology

Proposed violations of measurement independence face a substantive explanatory challenge. The required correlation must be accounted for even when settings are chosen through different physical processes, including human decisions, random-number generators, or astronomical signals. Simply calling the correlation a property of a complete history does not yet explain its magnitude, robustness, or physical origin.

Such proposals are sometimes described using terms including *measurement-dependent*, *retrocausal*, and *superdeterministic*. These labels are not interchangeable. Measurement dependence names the statistical failure of $P(\lambda|x,y)=P(\lambda)$. Retrocausal proposals posit a particular causal role for later boundary conditions or events; superdeterministic proposals generally allow the relevant setting–hidden-variable dependencies without adopting ordinary measurement independence. No causal mechanism follows from the conditional probability alone.

The description “conspiratorial” expresses a concern about whether the required correlations can be given a credible physical explanation; it does not by itself prove or disprove the model. The appropriate standard is a developed account, not a rhetorical label. The Block Universe interpretation motivates examining global correlations, but it does not remove the need to explain them physically if the proposal is advanced as a theory.

A.8 Requirements for a Complete Physical Theory

The construction proves an exact representational possibility for singlet measurements. To turn that additional measurement-dependent hidden-variable proposal into a complete underlying physical theory, further work would be required:

- **Physical specification.** Identify the relevant hidden variables and give a physically grounded account of their statistical relationship with the measurement settings. Shared causal ancestry or global boundary conditions may be investigated, but neither is established here as the explanation.
- **General formulation.** Account for quantum predictions beyond this particular singlet construction, including other states, measurements, interference phenomena, and the established statistical framework.
- **Consistency with established physical constraints.** Explain how the correlations fit relativistic causal structure and preserve the absence of operational superluminal signaling, rather than merely reproducing no-signaling marginals in an illustrative distribution.

These requirements apply to the development of a complete physical hidden-variable account, **not** to the manuscript's separate ontological interpretation of wavefunction collapse. As Section 4.8 explains, treating the wavefunction as a non-ontic statistical description can dissolve the traditional demand for a physical collapse process within that interpretation, without requiring a newly derived hidden-variable theory.

The Appendix therefore establishes a limited but precise mathematical conclusion: local response functions with measurement-dependent hidden-variable statistics can reproduce the full singlet joint probabilities. It does not determine that nature uses this mechanism, explain the origin of the quantum probability distribution, or establish which physically possible history is actual.

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Further Reading and Foundational Works

This manuscript presents a personal synthesis of ideas concerning spacetime, quantum mechanics, the direction of time, free will, and time travel. Its arguments were developed through independent reasoning and extensive discussions, rather than as a systematic review of the scientific and philosophical literature.

The works listed below provide foundational scientific results, historical context, and additional perspectives on the subjects discussed. They are offered as resources for readers wishing to explore these topics further. Their inclusion does not imply that the arguments presented in this manuscript were derived directly from these publications, nor that their authors endorse the interpretations advanced here.

Relativity, Spacetime, and the Block Universe

- **Albert Einstein (1905).** *On the Electrodynamics of Moving Bodies*. Foundational formulation of special relativity, including the relativity of simultaneity.
- **Hermann Minkowski (1908).** *Space and Time*. The four-dimensional spacetime picture developed from special relativity; linked text is an English translation.
- **Stanford Encyclopedia of Philosophy.** *Being and Becoming in Modern Physics*. An overview of the philosophical debate about time, becoming, and the implications of relativity.

Thermodynamics and the Direction of Time

- **Stanford Encyclopedia of Philosophy.** *Thermodynamic Asymmetry in Time*. Background on entropy, statistical mechanics, and why the thermodynamic arrow has a preferred direction.

Quantum Mechanics and Bell's Theorem

- **John S. Bell (1964).** *On the Einstein Podolsky Rosen Paradox*. The foundational inequality argument and the assumptions of local hidden-variable descriptions.
- **Michael J. W. Hall (2010).** *Local Deterministic Model of Singlet State Correlations Based on Relaxing Measurement Independence*. An example of prior research on setting-dependent hidden-variable distributions; the paper's model is deterministic and is not the model constructed in this manuscript.
- **Yoon-Ho Kim and colleagues (2000).** *Delayed "Choice" Quantum Eraser*. A foundational experiment involving delayed-choice arrangements and conditional detection statistics.

Free Will

- **Stanford Encyclopedia of Philosophy.** *Free Will*. An introduction to compatibilist, incompatibilist, libertarian, and related accounts of agency.

Time Travel and Causal Consistency

- **John Friedman and colleagues (1990).** *Cauchy Problem in Spacetimes with Closed Timelike Curves*. Research on the constraints imposed by closed timelike curves and globally consistent physical solutions.
- **Stephen W. Hawking (1992).** *Chronology Protection Conjecture*. A proposal concerning whether the laws of physics prevent the formation of closed timelike curves.